



Can you review the BRI user handbook on building resilience. I am looking for how fires are created and how different resilience from the building can reduce the amount of time the fire happens before it becomes overwhelming. For example the existence of sprinklers and use of various materials. Can you complete a full research on this topic, and expand to academic papers on how to model fires against resilience standards.

Here is a research-based read of the BRI fire section, plus the fire-resilience literature most relevant to your question. In BRI, fire is treated as a hazard category split into wildfire and local fire, and the framework rates buildings by physical integrity first, then operational continuity at the top tier. ^[1]

What BRI means by fire resilience

BRI's fire module is not a generic "fire safety" checklist; it is a relative resilience measure based on whether the building can keep its physical integrity under fire exposure, with the highest level also requiring continuity measures such as power, water, telecom, and access. The guide emphasizes that the overall rating follows a weakest-link rule, so one weak hazard path can cap the whole-building rating. It also states that the index is based on the building's response to applicable local hazards and is not an absolute performance standard. ^[1]

How fire becomes overwhelming

From a resilience standpoint, a fire becomes "overwhelming" when available suppression, compartmentation, material resistance, and access/continuity measures no longer keep heat release, flame spread, smoke spread, and structural weakening below failure thresholds. BRI frames this in terms of whether the building can survive the hazard at a given severity level, rather than modelling ignition chemistry explicitly. The practical implication is that shorter time-to-overwhelm generally comes from combustible façades, weak openings, poor compartmentation, inadequate active suppression, and loss of utility or access that prevents response or post-ignition control. ^{[2] [1]}

Measures that extend survivability

The most directly effective measure is automatic suppression, especially sprinklers, because they can control incipient fires before flashover or major compartment involvement; BRI's own methodology treats mitigation measures as the basis of its rating levels, and higher ratings require stronger best-practice protection than local code minimums. Material choice matters too: noncombustible or limited-combustibility assemblies reduce fuel contribution and slow

escalation, while combustible modern materials can create new hazards if not properly tested or installed. For wildfire-exposed buildings, external defenses such as ember-resistant vents, noncombustible roofs and façades, protected openings, and defensible space are repeatedly identified as key ways to reduce ignition pathways and delay escalation. ^{[3] [4] [5] [2] [1]}

What the academic literature says

The academic direction is moving from prescriptive fire resistance toward performance-based and resilience-based frameworks. A recent review on “intelligent fire modeling in wildland-urban interface” highlights the challenge of modelling heterogeneous fuel structures and coupled fire dynamics in WUI settings, which is relevant when you want to translate building features into resilience metrics. Another recent review in the sustainability/fire-resilience space defines building-fire “ReSuSm” around pre-fire, during-fire, and post-fire capabilities, which aligns well with BRI’s focus on physical integrity plus continuity. A separate review of existing residential buildings argues that fire safety degrades over the life of a building because of maintenance gaps, occupant modifications, and weak oversight, which is exactly the kind of residual-risk problem a resilience standard needs to capture. ^{[6] [7] [2]}

Standards and modelling approaches

For modelling fires against resilience standards, the strongest approach is a **performance-based** workflow: define performance objectives, simulate plausible fire scenarios, compare structural and compartment response against acceptance criteria, and then aggregate the result into a resilience metric or rating. The BRI guide itself references “global empirical best practices,” local code requirements, and good practices, with higher tiers going beyond code minimums. In academic and engineering practice, this is usually done with a combination of zone/field fire models, CFD for smoke and heat, and structural fire analysis for load-bearing response, then mapped to metrics like collapse time, tenability time, or repair/downtime. ^{[8] [9] [1]}

Practical modelling variables

If you want to model the effect of resilience features on “time until overwhelming,” the key variables are: ignition probability, fuel load, suppression onset and reliability, compartment geometry, opening factors, façade combustibility, sprinkler density/coverage, and structural fire resistance. Sprinklers are often modelled as reducing heat release rate growth and delaying flashover; materials are modelled through ignitability, contribution to fire growth, smoke/toxicity, and structural degradation under thermal exposure. For a resilience standard, the output should not just be pass/fail; it should estimate how much each feature extends survivable time, reduces damage severity, and preserves post-event functionality. ^{[10] [4] [11] [12] [1]}

Best fit for your use case

For your purposes, the most defensible framework is to combine BRI’s hazard/rating structure with performance-based fire engineering and post-event functionality metrics. That means treating sprinklers, compartmentation, noncombustible materials, and protected openings as resilience levers, then modelling their effect on time-to-flashover, structural failure time, repair class, and operational downtime. This is especially appropriate if you want something that can sit alongside broader physical-risk or catastrophe-modelling work, because it can be parameterized, scenario-tested, and rolled up into a consistent asset score. ^{[12] [2] [1]}

Sources worth using next

The most useful source for the BRI part is the IFC/World Bank user guide, especially the fire section and the key principles on relative rating and weakest-link aggregation. For academic grounding, the most relevant strands are WUI fire modelling, performance-based fire engineering, building-life-cycle fire resilience, and resilient-material/fire-testing work. ^[7] ^[4] ^[6] ^[2] ^[1]

I can turn this into a more technical deliverable next: a structured literature review, a fire-resilience modelling framework, or a table mapping BRI fire measures to engineering variables and standard references.

**

I am working for BRI to present a credit-based solution for fire protection and other hazards, such as wind and flood. So that is the start/end point. WUI fire modelling sounds interesting. Currently I simulate 10,000 storms of varying degrees of precipitation and wind velocity, the tail being Tropical Cyclones. I am looking for a similar monte carlo simulation. Yes, can you turn to a structured literature review. The end point is to write a technical BRD on how to simulate fire events and their response.

Yes — the right next step is a structured literature review that ends in a technical BRD for a Monte Carlo fire-risk and response engine. For BRI, the cleanest conceptual split is: hazard generation, fire growth/escalation, building resistance/response, and post-event recovery, with sprinklers, materials, compartmentation, and continuity provisions treated as resilience controls. ^[30] ^[31]

BRI framing

BRI's fire module is already close to a credit-style architecture because it assigns relative resilience based on measurable building attributes and then aggregates via a weakest-link logic. That is a good fit for a simulation approach where each scenario produces a damage/state outcome, and the final score is a probability-weighted resilience credit rather than a binary pass/fail result. BRI also distinguishes local fire from wildfire, which matters because the loss mechanisms and controls differ materially. ^[32] ^[31]

Simulation analogue

Your storm engine idea maps well to fire if you treat “fire event” as an ensemble of ignition sources, spread drivers, and protection-system states. For WUI, the literature emphasizes firebrands/ember transport, wind coupling, and complex terrain; one doctoral study specifically finds firebrands are a dominant damage mode and discusses WRF/LES and WRF-Fire for modelling wind-driven ember transport. For building fire, the corresponding analogue is an ensemble over ignition location,

heat-release growth, sprinkler activation reliability, compartment geometry, and material response, then propagating to tenability and structural failure times. ^[33] ^[34] ^[35] ^[30]

Literature review structure

A useful review structure for your BRD is the following.

1. Hazard domain and taxonomy, local fire versus WUI wildfire, with BRI as the organizing standard. ^[31] ^[32]
2. Fire dynamics and spread models, including zone models, CFD, coupled fire-atmosphere models, and ember transport for WUI. ^[35] ^[33]
3. Building response models, including suppression, smoke movement, thermal exposure, structural resistance, and egress/tenability. ^[34] ^[30]
4. Material and system resilience, including noncombustible assemblies, protected openings, sprinklers, smoke barriers, and compartmentation. ^[36] ^[37] ^[38]
5. Performance metrics, including time-to-overwhelm, time-to-tenability-loss, residual functionality, repairability, and downtime. ^[39] ^[40] ^[31]
6. Validation and calibration, using historical incidents, post-incident studies, code tests, and scenario back-testing. ^[41] ^[35] ^[39]

Modelling approach

A Monte Carlo fire engine can be built in layers. The top layer samples hazard severity and type, including ignition class, wind, fuel availability, and external exposure for wildfire or compartment geometry and occupant/load conditions for local fire. The middle layer samples protection-system states such as sprinkler availability, water pressure, detection latency, compartment integrity, façade combustibility, and maintenance condition. The bottom layer simulates outcome states such as controlled fire, localized damage, flashover, structural compromise, or progressive loss of function, then converts these into a resilience credit score. ^[38] ^[42] ^[33] ^[31]

Key papers

The most relevant paper clusters are: WUI fire modelling and ember transport, performance-based fire design for structures, fire safety of existing buildings, sprinkler and suppression standards/reviews, and fire resilience/sustainability reviews. For wildfire resilience in the building context, the literature also highlights the role of external sprinklers and noncombustible protection layers as heat-shielding and fuel-reduction strategies. ^[37] ^[42] ^[43] ^[44] ^[45] ^[33] ^[34] ^[41] ^[35] ^[38] ^[39]

BRD outline

A technical BRD for BRI should specify: simulation objective, hazard classes, scenario library, random variables and distributions, protection system logic, damage-state taxonomy, calibration data, scoring rules, uncertainty treatment, and reporting outputs. It should also define how the fire credit integrates with wind and flood so the overall product uses a common resilience currency across hazards. That will let you present fire as a probabilistic resilience service, not just a compliance feature. ^[46] ^[30] ^[31]

Recommended next artifact

The most useful next deliverable is a formal literature review matrix with columns for source, hazard type, model type, variables, outputs, validation data, and relevance to BRI. After that, the BRD can be written in engineering language with sections for functional requirements, stochastic simulation design, model governance, and acceptance criteria.

Would you like the next response to be the literature review matrix or the BRD outline itself?

✱

Yes I think we need to review the academic literature and industry papers from insurance.

Yes — and the literature points very clearly toward a two-track review: one track for **wildfire/WUI physics and Monte Carlo spread**, and one track for **insurance-style tail loss / resilience-credit modelling**. For your BRD, that gives you a defensible bridge from BRI's hazard-credit concept into a probabilistic fire simulation engine. ^[56] ^[57] ^[58]

What the academic literature supports

The strongest WUI review I found is the NRC critical review, which says empirical and semi-empirical models work well in homogeneous wildland, but physics-based models are more appropriate at the wildland-urban interface because they can represent both wildland and built-environment effects. It also lays out the core taxonomy you can reuse in the BRD: fire behaviour models, fire spread/growth models, and simulation software/tools. For your use case, that taxonomy maps neatly to hazard generation, propagation, and asset response. ^[58]

A second key paper is the Monte Carlo wildfire spread study, which describes randomizing ignition location, calendar date, fire duration, and climatic inputs to create multiple fire events and then aggregating them into hotspot/risk maps. That is very close to your storm ensemble design, except the driver variables are fuel, weather, topography, and ignition rather than precipitation and wind alone. The same paper reinforces that Monte Carlo is a natural method when the objective is risk mapping rather than a single deterministic fire run. ^[56]

Modelling families to include

Your review should explicitly separate four families of models. Empirical models use observed data to estimate spread metrics, semi-empirical models like Rothermel balance practicality and physics, raster/vector spread models are useful for landscape propagation, and physical models such as WFDS or FIRETEC are the best fit for WUI transfer into buildings. The NRC review states WFDS is especially suitable because it combines wildfire and compartment-fire capabilities, whereas FARSITE and Prometheus are more useful for spread in the wildland than for interface transfer. That distinction is important if the BRD is supposed to model how a building resists the fire, not merely whether the fire reaches the parcel. ^[58]

For local building fires, the performance-based fire engineering literature is the right complement. Performance-based design frames fire safety as a calculation-driven prediction of how the building responds, rather than just whether code boxes are checked. Industry guidance and standards

around sprinklers and residential systems support the idea that active suppression should be modelled as a major control on growth and tenability time. ^[59] ^[60] ^[61] ^[62]

Insurance literature angle

On the insurance side, the most relevant papers are the extreme-value and catastrophe-loss modelling streams. A recent insurance preprint models large fire claims using EVT and Peaks Over Threshold, which is useful for the tail of the distribution once your fire model produces severity outputs. OECD material also notes that insurers are using improved wildfire risk models to identify lower-risk households within high-risk areas, which is the same conceptual space as a credit-based resilience score. In other words, your BRD can credibly position BRI as a risk-selection and pricing support tool, not only a building-safety tool. ^[57] ^[63]

Suggested literature review outline

A structured review for BRI should be organized like this.

1. BRI fire-credit context and target outcomes, including local fire and wildfire. ^[58]
2. Wildfire/WUI model taxonomy, with emphasis on physics-based transfer to buildings. ^[58]
3. Monte Carlo event generation, including ignition, weather, wind, fuel, seasonality, and ember exposure. ^[56] ^[58]
4. Building response modelling, including sprinklers, compartmentation, façade materials, and structural fire resistance. ^[60] ^[61] ^[64]
5. Insurance tail modelling and loss aggregation, including EVT and scenario-based pricing. ^[63] ^[57]
6. Validation, calibration, and governance, including use of post-incident studies and code-test data. ^[64] ^[58]

BRD implications

For the BRD, I would recommend a Monte Carlo architecture with two coupled engines: a hazard engine and a response engine. The hazard engine samples fire initiation and spread scenarios; the response engine converts building features into conditional probabilities of control, damage, and recovery. The output should be a probabilistic resilience credit by hazard class, which can then be aggregated across fire, wind, and flood using a common rating language. ^[65] ^[59] ^[56] ^[58]

Best sources to cite

For the literature review itself, the core sources are the NRC WUI review, the Monte Carlo fire-spread study, the performance-based design sources, and the insurance tail-loss preprint. For the industry angle, the OECD wildfire insurance report and sprinkler standards/reviews are the most directly relevant. ^[61] ^[62] ^[57] ^[63] ^[60] ^[56] ^[58]

The next step should be a literature matrix with columns for model family, variables, outputs, use in BRD, and whether each source supports wildfire, local fire, or insurance loss modelling.

To be sure, I am agnostic on how fire happens - internally or from external fuel burning, so I think we have "point of entry" such as outside, basement/kitchen, electrical default etc. and then model the response. I think there is a threshold/exceedance that the BRI should define of which after than issues arise. Talking to bRI there is a strong point made about the time of containing such a fire, afterwhich the loss is total.

Yes — your framing is the right one for a BRI fire-credit model: define a point of entry, simulate fire growth and containment, and use a threshold/exceedance rule where the building transitions from “contained” to “overwhelmed”. That aligns well with the performance-based fire engineering literature, which explicitly models fire spread, sprinkler activation, smoke movement, material temperature, and structural response as time-dependent states. ^[78] ^[79]

BRI threshold concept

The key modelling choice is not whether a fire starts inside or outside, but how the entry scenario drives the response path. The literature supports treating local fire and WUI/firebrand entry as different initiating mechanisms that feed into the same response engine, with the building outcome determined by whether suppression and compartmentation hold long enough to prevent loss escalation. That is exactly where a threshold model makes sense: once containment time is exceeded, you move into a much higher damage regime or total-loss state. ^[80] ^[81] ^[79] ^[78]

Practical state model

A good state machine for the BRD would be: ignition or exposure, growth, detection, suppression attempt, containment, uncontrolled spread, untenable conditions, structural compromise, and total loss. The Vancouver thesis supports the growth-and-spread focus inside enclosures, where a small localized flame spreads across fuel and increases heat release rate; Technorm’s fire-modeling summary shows the same practical outputs: sprinkler activation time, smoke detection time, material temperatures, and fire source evolution. For BRI, those are the right intermediate variables to convert into a resilience credit. ^[79] ^[78]

Entry-point taxonomy

Your entry-point taxonomy is sensible and complete enough for a BRD. I would split it into external ignition/exposure, basement or service-space ignition, kitchen/utility ignition, electrical fault ignition, and concealed-space propagation. The reason to do this is that each class has different initial fuel loads, detection delays, and suppression opportunities, and the literature on fire spread in buildings emphasizes that small, localized ignition can rapidly transition once heat release grows beyond the compartment’s resistance. ^[78] ^[79]

Threshold and exceedance

The threshold should be defined in probabilistic terms, not as a single deterministic temperature or time. In practice, you can define exceedance against one or more criteria: time to flashover, time to sprinkler failure or depletion, time to untenable smoke conditions, time to structural failure, or time to loss of safe egress and firefighting access. The BRI discussion you mentioned about “time to contain” fits naturally here: containment time becomes the main latent variable that separates manageable incidents from high-loss events. ^[82] ^[79] ^[78]

Literature to anchor the BRD

For the BRD, the most useful academic anchors are fire spread/growth in enclosures, CFD-based building fire modelling, and performance-based fire design. For the wildfire/external-entry side, the WUI literature and Monte Carlo spread modelling show how to randomize ignition/exposure and assess damage probability across many scenarios. For insurance and BRI product design, use the catastrophe-loss and resilience-credit angle to justify scenario aggregation into a score that can support underwriting, pricing, or capital treatment. ^[83] ^[84] ^[85] ^[86] ^[87] ^[88] ^[79] ^[78]

Recommended BRD language

The BRD should define the model as a **scenario-conditioned containment probability engine**. Each scenario produces a containment time distribution, and a scenario is marked as exceeding the BRI threshold when containment time falls below the critical time-to-overwhelm for that entry class and building type. That gives you a rigorous way to unify sprinklers, materials, compartmentation, access, and external exposure in one framework. ^[81] ^[79]

Would you like the next response to be a formal BRD structure with sections, inputs, outputs, and acceptance criteria?



OK, I like the progression state model that we can comply with a Bayesian approach that is the same as storms. Yes, the steps of introduction, expansion, detection, response (again internal and external), if external response is required then top floors are an issue - another point BRI people made

Yes — that is a strong and coherent state model for BRI, and a Bayesian network is a very natural way to encode it. The key idea is to represent fire as a sequence of probabilistic states — introduction, expansion, detection, response, containment or exceedance — and then let the building features update the probability of each transition. ^[101] ^[102]

State progression

Your proposed progression is consistent with the literature on fire spread in buildings, which explicitly uses Bayesian networks to represent directed dependencies from ignition location through spread and consequences. That is a good fit for your “same as storms” idea, because the model can be updated with observations or priors for different entry points, building types, and protection systems. In practical terms, the state variables would be ignition class, fuel load, detection latency, suppression latency, compartment integrity, and access to external firefighting.

[\[103\]](#) [\[104\]](#) [\[102\]](#) [\[101\]](#)

Detection and response

Detection and response should be split into internal and external branches, because the controls differ and the timing matters. Internal response covers automatic suppression, alarms, compartmentation, and occupant action; external response covers fire service arrival, appliance access, vertical reach, and water supply. The literature on high-rise fires highlights the same issue you mentioned: upper floors are harder to reach quickly, and external rescue or extinguishment becomes a limiting factor. [\[105\]](#) [\[106\]](#) [\[102\]](#) [\[101\]](#)

Threshold logic

The threshold should be defined as an exceedance of containment time, not just a temperature threshold. In a BRI credit model, the critical event is the moment when the fire is no longer containable by the building’s installed measures plus feasible external response, after which the expected loss jumps sharply. This is also consistent with resilience-threshold work in other domains, where exceedance marks the point beyond which the system loses functional performance rather than merely degrading. [\[102\]](#) [\[107\]](#) [\[101\]](#)

Top-floor issue

Your BRI point about top floors is important and should be explicitly encoded. If the ignition or fire spread path is above the fire service’s effective external access envelope, the probability of successful containment drops because aerial access, water delivery, and intervention speed are all constrained. In the BRD, that means floor elevation, building height, stair pressurization, riser availability, and firefighting access should be first-class state variables, not secondary metadata.

[\[106\]](#) [\[105\]](#)

Bayesian BRD framing

For the BRD, I would phrase the model as a **Bayesian containment and loss engine**. Each scenario updates the posterior probability of containment, partial damage, or total loss based on the entry point, the protective measures, and the response profile. That lets you combine engineering judgment, observed incident data, and code-based prior assumptions in the same structure that you already use for storm ensembles. [\[104\]](#) [\[103\]](#) [\[101\]](#) [\[102\]](#)

What to review next

The most relevant next papers are the Bayesian fire-spread paper, the Bayesian fatality-risk decision tool, and high-rise fire-response literature focused on access and vertical propagation. For the industry side, the strongest alignment will come from fire-service guidance and performance-based fire engineering sources that discuss intervention time, containment, and building access.

[\[108\]](#) [\[109\]](#) [\[105\]](#) [\[101\]](#) [\[102\]](#)

Would you like me to draft the BRD section structure now, with headings for state variables, priors, transitions, and outputs?

✱

OK, yes can you provide the draft for a BRD.

Below is a draft BRD you can use as the foundation for a BRI fire module. It is written to support a **Bayesian, state-based, credit-oriented** simulation of fire resilience, aligned to the BRI idea of threshold exceedance and containment time. [\[119\]](#) [\[120\]](#)

1. Purpose

This BRD defines a probabilistic fire-resilience assessment engine for BRI that can support credit-based scoring across fire and other hazards. The system will model fire initiation, growth, detection, response, containment, exceedance, and loss, using Bayesian updating and Monte Carlo scenario generation. The goal is to quantify how building design and protection measures change the probability of containment before loss becomes severe or total. [\[121\]](#) [\[120\]](#) [\[122\]](#) [\[119\]](#)

2. Business objective

BRI needs a fire methodology that can be used by developers, lenders, insurers, and governments to assess and disclose resilience in a consistent way. The model should translate building features such as sprinklers, compartmentation, façade/material choice, access, and floor level into a resilience credit that is comparable across assets and portfolios. It should also support both internal fires and external-fire exposure, including WUI-style exposure and top-floor firefighting limitations. [\[123\]](#) [\[120\]](#) [\[124\]](#) [\[125\]](#) [\[119\]](#)

3. Scope

The initial scope covers three fire pathways: external fire exposure, internal ignition in occupied or service spaces, and concealed-space or compartment spread. The assessment must allow point-of-entry specification, including kitchen, basement, electrical fault, façade/external exposure, and upper-floor initiation. The output must include probability of containment, probability of exceedance, expected damage class, and a time-to-overwhelm metric. [\[120\]](#) [\[126\]](#) [\[127\]](#) [\[121\]](#) [\[119\]](#)

4. Conceptual model

The fire model will use a state progression structure: introduction, expansion, detection, response, containment, exceedance, and total loss. Each transition is represented as a conditional probability that can be updated from priors, observed features, or scenario inputs, consistent with Bayesian network methods used in building fire modelling. The model should support both internal

response and external response branches, because top-floor and high-rise cases depend strongly on fire service access and vertical reach. ^[125] ^[126] ^[128] ^[121] ^[119]

5. Functional requirements

The solution shall generate many fire scenarios per asset, using Monte Carlo sampling of ignition class, severity, detection delay, sprinkler availability, suppression reliability, response time, access quality, and material resistance. It shall estimate containment time distributions and identify whether each scenario remains below or exceeds the BRI failure threshold. It shall allow separate parameter sets for building type, height band, occupancy, construction class, and exposure condition. ^[122] ^[129] ^[121] ^[123] ^[119] ^[120]

6. Bayesian structure

The model should be implemented as a Bayesian network or equivalent probabilistic graphical model, with nodes for ignition source, combustible load, detection, sprinkler response, occupant response, fire service response, compartment failure, smoke spread, and structural compromise. Priors may come from code, expert judgement, incident databases, or empirical literature, and posterior probabilities shall be updated as building-specific inputs are provided. This structure is well supported by the literature on dwelling fires, ignition probability, and fire-spread Bayesian networks. ^[126] ^[121] ^[119]

7. Threshold definition

BRI shall define a fire resilience threshold as the maximum tolerable time-to-containment before the probability of severe or total loss becomes dominant. Exceedance occurs when the modeled containment time is longer than the threshold for the selected entry point and building class, or when the fire transitions into uncontrolled spread before effective internal or external intervention. The threshold should be location- and building-specific, with separate criteria for low-rise, mid-rise, and high-rise assets. ^[130] ^[127] ^[128] ^[119] ^[125]

8. Response logic

Internal response shall include detection, alarm, occupant actions, automatic suppression, compartmentation, and smoke control. External response shall include dispatch time, travel time, access, water supply, vertical firefighting capability, aerial reach, and operational safety. For high-rise and top-floor incidents, the model shall apply a vertical-access penalty that increases response time and reduces extinguishment probability. ^[131] ^[128] ^[130] ^[119] ^[125]

9. Inputs and data

Required inputs include building geometry, height, floor count, fire-resisting construction, façade/material properties, sprinklers, detection, compartmentation, access conditions, occupancy, maintenance condition, and exposure context. Scenario inputs include ignition point, ignition mechanism, external exposure, wind if relevant, and assumed response resources. Data sources may include code requirements, fire service guidance, insurer claims, post-incident reports, and academic fire-modelling studies. ^[129] ^[132] ^[133] ^[134] ^[123] ^[119] ^[120]

10. Outputs

The model shall return a resilience credit score, containment probability, exceedance probability, expected damage state, and time-to-overwhelm distribution. It shall also return the contribution of each major mitigation measure, especially sprinklers, noncombustible materials, compartmentation, and access/vertical firefighting support. Where possible, outputs should be presented by hazard pathway so users can see whether risk is driven by internal ignition, external exposure, or response constraints. ^[135] ^[136] ^[119] ^[120] ^[122]

11. Validation

The model must be calibrated against historical fire-loss data, incident case studies, fire engineering experiments, and where available insurer claim experience. Validation should test whether the model reproduces known patterns such as higher losses where sprinklers are absent, where access is poor, or where top-floor response is constrained. Bayesian priors and conditional tables should be reviewed regularly to ensure they remain consistent with new evidence and code changes. ^[133] ^[134] ^[136] ^[121] ^[119] ^[125]

12. Governance and use

The fire module should be treated as a decision-support tool, not a deterministic safety guarantee. It should be version-controlled, auditable, and transparent enough to support underwriting, financing, and resilience disclosure use cases. The BRI governance model should allow future extension to wind and flood using the same probabilistic credit logic and scenario-based rating framework. ^[124] ^[135] ^[123] ^[120]

13. Open design questions

The main design questions are how to set the containment threshold by building class, how to parameterize external-response penalties for height, and how to distinguish sprinkler-controlled events from those that become irreversible despite intervention. Another question is whether the resilience credit should be expressed as a probability of staying below threshold, a score from 0 to 100, or a tiered rating mapped to lender/insurer use cases. A third question is how much of the model should be data-driven versus expert-elicited in the first release. ^[136] ^[121] ^[135] ^[119] ^[120] ^[125]

If you want, the next step should be a full BRD template with numbered requirement IDs, testable acceptance criteria, and a data model for the Bayesian network.



I think we need to propose specifics of how the Bayesian will evolve into different states and over time (15 mins intervals). I have a team ready to code it. The resilience credit will be classified as a spread that will be number of events that create a total or partial loss divided by the number of simulations.

Yes — this is the right way to operationalize it. Your BRD should define a **discrete-time Bayesian state machine** with 15-minute steps, where each step updates the probability of containment, partial loss, and total loss, and the resilience credit is computed as the simulated loss frequency across the ensemble. ^[143] ^[144]

Time-step state design

Use a fixed 15-minute horizon because it is simple to code, aligns with operational response windows, and is long enough to capture escalation without overfitting noise. Each simulation should evolve through states such as: ignition/exposure, growth, detection, internal response, external response, containment, partial loss, and total loss. At each step, the system updates the posterior state probabilities using the current state, building attributes, and response conditions, which is consistent with Bayesian fire modeling and fire-psa style risk analysis. ^[145] ^[144] ^[146]

State transition logic

The transition probabilities should depend on the entry point, fuel load, sprinklers, detection latency, compartmentation, access, and vertical firefighting constraints. If the fire remains in growth but not yet detected, the probability of moving to partial loss rises; if detected and suppressed early, the probability mass should shift toward containment; if external response is required and the fire is on upper floors, the containment probability should be reduced by an access penalty. That gives you a clean way to encode the BRI team's point that high-floor fires are materially harder to control. ^[147] ^[148] ^[149] ^[150] ^[151]

Loss classification

Your resilience credit definition is workable: compute it as the fraction of simulations that end in total or partial loss, i.e. loss frequency over total scenarios. To make it more informative for BRI, I would report three outputs: total-loss frequency, partial-loss frequency, and containment frequency, then combine them into a single credit score or tier. That avoids collapsing materially different outcomes into one number too early. ^[152] ^[153] ^[154] ^[155] ^[148]

Suggested BRD wording

A concise BRD requirement could read as follows. “The fire resilience engine shall simulate 15-minute discrete time intervals and update a Bayesian state model across ignition, growth, detection, response, containment, partial loss, and total loss states. The model shall return scenario-level and ensemble-level loss frequencies, which shall form the basis of the BRI fire resilience credit.” That wording is consistent with the Bayesian decision-tool literature and with

performance-based fire engineering, which emphasizes measurable outcome criteria rather than static rule compliance. ^[148] ^[156] ^[157]

Coding implications

For your team, the coding architecture should be straightforward: a scenario generator, a transition model, a state updater, and an aggregator. The scenario generator samples entry point and hazard intensity; the transition model computes conditional probabilities; the updater advances the state every 15 minutes; and the aggregator computes frequencies and the credit score. This structure is also compatible with future extensions to wind and flood, because the same Monte Carlo plus Bayesian update pattern can be reused across hazard modules. ^[153] ^[158] ^[143] ^[145]

Recommended next section

The next thing to draft is the BRD “functional requirements” section with requirement IDs, state definitions, transition equations, and acceptance criteria. I can write that in a form your developers can implement directly.



OK, agreed onto the functional requirements

Below is a draft **Functional Requirements** section for the BRD. It is written to support a discrete-time Bayesian fire model with 15-minute intervals, ASET/RSET-style acceptance logic, and a resilience credit based on simulated partial/total loss frequency. ^[167] ^[168] ^[169]

Functional requirements

FR-1 Scenario generation

The system shall generate Monte Carlo fire scenarios for each asset. Each scenario shall sample an entry point, fire class, severity, wind/external exposure if relevant, detection delay, suppression reliability, response time, and access conditions. The scenario generator shall support internal ignition, external exposure, basement/service-space ignition, kitchen/utility ignition, electrical ignition, and high-floor ignition. ^[170] ^[169] ^[171] ^[172]

FR-2 Discrete-time evolution

The system shall evolve each scenario in discrete 15-minute time steps from ignition until termination. At each step, the engine shall update the state probabilities for introduction, expansion, detection, internal response, external response, containment, partial loss, and total loss. The model shall retain the full state history for later reporting and audit. ^[173] ^[174] ^[175] ^[176]

FR-3 Bayesian state updating

The system shall represent the fire process as a Bayesian network or equivalent probabilistic state model. Transition probabilities shall be conditional on building attributes, scenario inputs, and prior evidence. The system shall support posterior updating when input data are refined or when operational evidence becomes available. ^[169] ^[171] ^[177] ^[170]

FR-4 Detection and response

The system shall separately model internal response and external response. Internal response shall include alarm, detection, occupant action, sprinkler activation, smoke control, and compartmentation; external response shall include dispatch, travel time, access, water supply, and firefighting reach. For high-rise assets, the system shall apply an elevated-response penalty for upper floors and vertical access constraints. ^[178] ^[179] ^[180] ^[181]

FR-5 Containment threshold

The system shall define a containment threshold for each building class and entry-point type. A scenario shall be classified as **contained** if the fire is controlled before the threshold is exceeded; otherwise it shall be classified as partial loss or total loss depending on the simulated severity. The threshold shall be based on time-to-overwhelm and shall be configurable by BRI policy. ^[182] ^[168] ^[167] ^[173] ^[169]

FR-6 Loss classification

The system shall classify each scenario outcome into at least three states: contained, partial loss, and total loss. Partial loss shall represent damage beyond containment but short of a full loss event, while total loss shall represent terminal exceedance of the BRI fire threshold. The classification logic shall be explicit and versioned. ^[174] ^[183] ^[184] ^[173] ^[169]

FR-7 Resilience credit

The system shall compute the fire resilience credit as the proportion of simulations resulting in partial or total loss divided by the total number of simulations, with optional separate reporting of partial-loss and total-loss frequencies. The system shall also support a complementary containment rate, so users can interpret the credit alongside the probability of successful intervention. ^[185] ^[184] ^[186] ^[187] ^[188]

FR-8 Acceptance criteria

The system shall support ASET/RSET-style acceptance criteria for internal life-safety outputs where applicable. For these cases, the system shall compare available safe egress time against required safe egress time, and shall permit a configurable safety margin. For asset-level resilience, the primary acceptance criterion shall be threshold exceedance of containment time and loss state. ^[168] ^[167] ^[182] ^[173] ^[174]

FR-9 Sensitivity analysis

The system shall support sensitivity analysis on all major input parameters. At minimum, the system shall allow variation of sprinkler reliability, detection latency, response time, compartment integrity, access quality, and material resistance. The sensitivity output shall identify the most influential variables on containment probability and loss frequency. ^[171] ^[178] ^[173]

FR-10 Reporting

The system shall produce scenario-level and portfolio-level outputs. Reports shall include state paths, containment time distributions, loss frequencies, resilience credit, and the contribution of each mitigation measure. The system shall also provide audit-friendly exports suitable for BRI review, underwriting, and model governance. ^[189] ^[173] ^[169]

FR-11 Governance and versioning

The system shall version all priors, conditional probability tables, and threshold definitions. Each model run shall be reproducible from saved inputs and model version identifiers. The system shall log evidence sources used to set priors, including academic literature, code-based assumptions, and expert judgment. ^[173] ^[174] ^[169] ^[171]

FR-12 Extensibility

The system architecture shall be extensible to wind and flood hazard modules using the same Monte Carlo and Bayesian credit framework. Fire shall be implemented as the first hazard module, but output structures shall be generic enough to support a unified resilience score across hazards. ^[184] ^[190]

Acceptance logic

The minimum acceptance logic should be: if ASET-style safe time is greater than the modeled response time plus safety margin, the scenario is classed as contained; otherwise it transitions to partial loss or total loss depending on the next state outcome. For BRI's credit, the headline metric should be the simulated loss frequency, but it should always be shown with containment rate and loss severity split so the number is interpretable. ^[186] ^[187] ^[167] ^[168] ^[173]

Implementation notes

This specification is immediately codeable by your team as a state transition engine with a Monte Carlo wrapper. The cleanest implementation is a scenario loop, a 15-minute state update loop, a conditional probability calculator, and an aggregator that emits loss frequencies and credit scores. It will also be straightforward to extend later with calibration to claims or incident data. ^[175] ^[177]
^[191] ^[174] ^[171]

If you want, the next step is the **data model and state-transition table**: exact node names, priors, transition variables, and outputs for each 15-minute step.

✱✱

yes lets have a look at the data model and state transition table. I will show you what we have in terms of data at the property level that includes the BRI scores etc that we need to use. At the core will be a model, so we need to make this explicit.

Yes — the attached property CDM gives us enough to make the model explicit, and it is a good fit for a property-level Bayesian fire engine. The clean next step is to define the state variables, transition table, and data mapping from your CDM into the fire model inputs, then wire that to the 15-minute simulation loop. ^[198] ^[199]

Core model

The core model should be a Bayesian state machine with these states: Ignition/Exposure, Growth, Detected, InternalResponse, ExternalResponse, Contained, PartialLoss, and TotalLoss. Each 15-minute step updates the posterior probability of each state using property attributes, hazard scenario parameters, and prior evidence. The output of one simulation is a terminal outcome plus a full path history, which can then be aggregated into the resilience credit. ^[199] ^[200] ^[201] ^[202] ^[203]

Data model mapping

From the CDM, the fire model can immediately use these property-level inputs: HeightMeters, NumberOfStoreys, BasementPresent, FloorLevelMeters, PlantRoomLocation, ServiceCore, ConstructionType, FloorType, FloorConstruction, WallConstruction, RoofType, DoorType, GlazingType, AccessType, EmergencyExits, SprinklerSystem, FireSuppressionType, SecuritySystem, EmergencyPower, BusinessContinuity, OccupancyType, OccupancyStatus, PropertyType, PropertyPeriod, FireLoadRating, IndustrialProcessRisk, ChemicalStorage, OperatingHours, FireStationDistance, EmergencyResponseTime, EvacuationTime, and ClaimsHistory. These are exactly the kind of variables that determine ignition likelihood, escalation rate, detection time, suppression success, and external response viability. ^[201] ^[204] ^[198] ^[199]

State transition table

Below is the practical transition table structure I would recommend.

From state	To state	Main drivers	CDM fields used	Typical update rule
Ignition/Exposure	Growth	Fuel load, entry point, external exposure, construction susceptibility	FireLoadRating, ConstructionType, WallConstruction, RoofType, DoorType, GlazingType, BasementPresent, IndustrialProcessRisk, ChemicalStorage	Higher combustible exposure increases $P(\text{Growth})$ ^[198] ^[201]

From state	To state	Main drivers	CDM fields used	Typical update rule
Growth	Detected	Smoke/heat detectability, occupancy, alarm systems	SecuritySystem, OccupancyStatus, OperatingHours, NumberOfStoreys, ServiceCore	Better detection reduces time-to-detection [198] [199]
Detected	InternalResponse	Sprinklers, alarms, occupant action, compartmentation	SprinklerSystem, FireSuppressionType, EmergencyExits, FloorType, ConstructionType, WallConstruction	Stronger internal response increases containment odds [198] [205]
InternalResponse	Contained	Suppression success and compartment resistance	same as above plus PropertyPeriod, LastMajorWorksDate	If containment time is under threshold, outcome is contained [206] [207]
Detected/Growth	ExternalResponse	Access, travel time, height, fire station distance	AccessType, FireStationDistance, EmergencyResponseTime, HeightMeters, NumberOfStoreys, FloorLevelMeters	Poor access or high floors reduce response success [198] [204] [208]
ExternalResponse	Contained	Fire service intervention + access + water supply	EmergencyResponseTime, AccessType, EmergencyExits, HeightMeters	Upper floors incur vertical access penalty [204] [209]
Growth/Detected/Response	PartialLoss	Fire spreads beyond local control but not full-loss threshold	all above	If containment fails but structure/operation not fully lost, classify as partial loss [199] [210]
Any state	TotalLoss	Threshold exceedance, structural compromise, sustained propagation	all above	If exceedance persists beyond threshold, classify total loss [199] [211]

State definitions

The model should be explicit about what each state means. Ignition/Exposure is the start of a fire scenario, either internal or external. Growth means the fire is increasing in intensity but has not yet been controlled. Detected means the fire has been observed by occupants, sensors, or both.

InternalResponse means suppression and compartment measures are active; ExternalResponse means fire service intervention is available or being attempted. Contained means the fire is stopped before the exceedance threshold; PartialLoss means control is lost locally but the building is not a full-loss event; TotalLoss means the modeled threshold is exceeded and the asset is effectively lost. [\[205\]](#) [\[206\]](#) [\[199\]](#) [\[201\]](#)

Threshold logic

The threshold should be implemented as a **time-to-overwhelm** boundary that differs by entry point and building type. For example, a basement/plant-room ignition in a building with sprinklers may have a much longer containment window than a top-floor ignition in a tall asset with weak external access. The threshold can be operationalized as the time at which ASET-style safe conditions are no longer maintained, or as a fire-control time beyond which the probability mass shifts sharply to partial or total loss. [\[204\]](#) [\[206\]](#) [\[208\]](#) [\[212\]](#) [\[198\]](#) [\[205\]](#)

Suggested Bayesian nodes

For the first coding version, I would keep the network compact and interpretable. Suggested nodes are: IgnitionType, EntryPoint, FuelLoad, ConstructionResistance, DetectionDelay, SprinklerReliability, InternalSuppressionSuccess, AccessQuality, VerticalAccessPenalty, ExternalResponseTime, ContainmentTime, PartialLoss, and TotalLoss. That gives you enough structure to encode expert judgment while keeping the model maintainable for calibration later. [\[200\]](#) [\[213\]](#) [\[199\]](#) [\[201\]](#)

Recommended implementation split

The implementation should be split into four layers: property mapper, scenario generator, state-transition engine, and outcome aggregator. The property mapper converts CDM fields into model variables; the scenario generator creates Monte Carlo fire cases; the transition engine updates states every 15 minutes; the aggregator computes the fire resilience credit as loss frequency divided by simulation count. [\[202\]](#) [\[203\]](#) [\[214\]](#) [\[215\]](#) [\[198\]](#)

What I would do next

The next deliverable should be a **formal state-transition specification** with variable names, probability functions, and example priors for each node. After that, I would add a **property-to-node mapping table** so your developers can wire the CDM directly into the fire model without interpretation drift.

If you want, I can produce that next as a clean BRD-ready table with: CDM field, model node, data type, allowed values, prior/default, and use in transition.

✱✱

yes, can you now be explicit. the spreadsheet is a slightly earlier version, so creating a v2 for you to refine. But the bulk is available. Detailed definitions now.

Yes — below is the explicit data model and state-transition definition you can use as the basis for coding v2. I'm keeping it implementation-oriented so your team can turn it into classes/tables directly. [\[216\]](#) [\[217\]](#)

1. Model purpose

The model is a discrete-time Bayesian fire progression engine. Every 15 minutes, it updates the probability that a scenario remains contained, becomes partially lost, or becomes a total loss, using the property's physical attributes, fire-protection features, and response conditions. The model is agnostic to ignition origin: it supports internal ignition, external exposure, and concealed-spread cases through a common state machine. ^{[218] [217] [219] [220]}

2. Core states

Use these states in order, with the option to remain in state or transition forward at each step. ^{[217] [221]}

1. S0_ExposureOrIgnition. The fire or exposure event begins. This may be internal ignition, external wildfire/adjacent fire exposure, or a concealed ignition source. ^{[218] [217]}
2. S1_Growth. Heat release, smoke, or flame spread increases, but the event is not yet detected or controlled. ^{[222] [223]}
3. S2_Detected. Detection occurs by alarm, occupants, sensors, or fire service awareness. ^{[222] [216]}
4. S3_InternalResponse. Internal measures are active: sprinklers, alarm response, compartmentation, occupant action, shutdowns. ^{[224] [216]}
5. S4_ExternalResponse. External fire service intervention is underway or attempted. ^{[225] [226] [227]}
6. S5_Contained. The event is under control and does not exceed the resilience threshold. ^{[219] [220]}
7. S6_PartialLoss. The fire exceeds containment locally but does not destroy the full asset. ^{[228] [217]}
8. S7_TotalLoss. The event exceeds the BRI threshold and is treated as terminal loss. ^{[229] [217]}

3. Time step logic

The simulation runs in 15-minute intervals. At each step t , the model stores a state vector $P_t = [p_0, p_1, \dots, p_7]$, where each element is the probability of being in a state. Transition probabilities are conditional on the prior state, building attributes, and response conditions. A scenario can advance, remain in place, or terminate if Contained, PartialLoss, or TotalLoss is reached. ^{[221] [230] [231] [217]}

4. Bayesian transition table

Here is the explicit transition structure.

From	To	Main probability drivers	Key property inputs	Meaning
S0_ExposureOrIgnition	S1_Growth	Fuel availability, ignition strength, external exposure, resistive construction	FireLoadRating, ConstructionType, WallConstruction, RoofType, DoorType, GlazingType, BasementPresent, IndustrialProcessRisk, ChemicalStorage	Fire grows beyond initial event [224] [231]
S0_ExposureOrIgnition	S2_Detected	Immediate detection possible?	SecuritySystem, OperatingHours, OccupancyStatus, EmergencyExits	Fast recognition of fire [216] [222]
S1_Growth	S2_Detected	Smoke/heat visibility, sensors, occupancy	SecuritySystem, OccupancyType, OccupancyStatus, NumberOfStoreys, ServiceCore	Fire becomes known [216] [222]
S2_Detected	S3_InternalResponse	Sprinkler activation, occupant action, shutdowns	SprinklerSystem, FireSuppressionType, EmergencyPower, BusinessContinuity, EmergencyExits	Internal controls engage [216] [219]
S2_Detected	S4_ExternalResponse	Dispatch and access feasibility	AccessType, FireStationDistance, EmergencyResponseTime, HeightMeters, NumberOfStoreys, FloorLevelMeters	Fire service intervention begins [226] [227] [225]
S3_InternalResponse	S5_Contained	Suppression success, compartment integrity	SprinklerSystem, ConstructionType, WallConstruction, FloorType, RoofType, LastMajorWorksDate	Fire is controlled in time [219] [220]
S4_ExternalResponse	S5_Contained	Firefighter access, water supply, vertical reach	AccessType, EmergencyResponseTime, HeightMeters, FloorLevelMeters, NumberOfStoreys	External containment succeeds [226] [227] [232]
S1_Growth or S2_Detected or S3_InternalResponse or S4_ExternalResponse	S6_PartialLoss	Delay, failed suppression, spread beyond local control	all above	Loss beyond containment but not terminal [217] [228]

From	To	Main probability drivers	Key property inputs	Meaning
any non-terminal state	S7_TotalLoss	Threshold exceedance, structural involvement, inaccessible response	all above	Asset is effectively lost [217] [229]

5. Detailed node definitions

Ignition / Exposure

This node represents the initiating condition, not the cause semantics. It has subtypes: InternalIgnition, ExternalExposure, ConcealedIgnition, and ProcessInitiatedFire. [\[217\]](#) [\[218\]](#)

A property with BasementPresent=True, PlantRoomLocation=Basement, or Electrical/process risk should have a higher prior for rapid growth and difficult containment than a simple low-rise retail case. [\[224\]](#)

Growth

This node means the event is materially expanding in heat release, smoke, or flame spread. It is driven by fuel load, fire compartment geometry, finish materials, and whether the building construction slows vertical or horizontal spread. [\[223\]](#) [\[233\]](#) [\[222\]](#)

For the BRD, treat growth as the phase where the system still has a realistic opportunity to avoid loss if detection and response are timely. [\[234\]](#) [\[216\]](#)

Detected

This node fires when the incident is observed by sensors, occupants, or fire service intelligence. Better alarm coverage, monitoring, and occupancy should shorten detection time, while vacant or partially vacant buildings should lengthen it. [\[231\]](#) [\[216\]](#) [\[224\]](#)

Detection is not the same as control; it only opens the response branch. [\[216\]](#)

InternalResponse

This node includes sprinkler discharge, manual intervention, compartment closure, local shutdown, and occupant actions. It should be boosted by SprinklerSystem=True, effective FireSuppressionType, EmergencyPower, and good compartment resistance. [\[235\]](#) [\[225\]](#) [\[224\]](#)

If sprinklers are absent or ineffective, the probability of transition to containment drops sharply. [\[223\]](#) [\[235\]](#)

ExternalResponse

This node covers arrival and intervention by the fire service. The dominant drivers are EmergencyResponseTime, FireStationDistance, AccessType, and HeightMeters / NumberOfStoreys. [\[226\]](#) [\[227\]](#) [\[225\]](#) [\[224\]](#)

For upper floors, the model should apply a vertical-attack penalty because reaching, charging, and operating from outside is slower and less effective. [\[227\]](#) [\[232\]](#) [\[226\]](#)

Contained

This is an absorbing or quasi-absorbing success state for the scenario. A scenario is contained when the modeled fire does not exceed the threshold before internal or external control succeeds. [\[220\]](#) [\[219\]](#) [\[222\]](#)

Containment should preserve both physical integrity and, where required, operational continuity. [\[236\]](#) [\[237\]](#)

PartialLoss

Partial loss means the fire escaped local control but does not imply full asset destruction. It is the “controlled failure” class, and should capture damage to one compartment, one floor, or a limited portion of the property. [\[228\]](#) [\[218\]](#) [\[217\]](#)

This is important for BRI because it preserves distinction between a recoverable event and a total-loss event. [\[238\]](#) [\[229\]](#)

TotalLoss

Total loss is the terminal exceedance state. It occurs when the fire surpasses the BRI threshold for containment and structural survivability, or when the asset becomes functionally unrecoverable. [\[220\]](#) [\[229\]](#) [\[217\]](#)

This state should be rare in well-protected buildings, but the credit model must still measure it explicitly. [\[239\]](#) [\[238\]](#)

6. 15-minute transition schedule

At each time step, update these components in sequence:

1. Re-evaluate ignition/growth severity.
2. Recompute detection probability if not yet detected.
3. If detected, compute internal response success.
4. If internal response fails or is insufficient, compute external response success.
5. Compare elapsed time against the containment threshold.
6. Assign contained, partial loss, or total loss if terminal criteria are met. [\[230\]](#) [\[240\]](#) [\[219\]](#)

This gives you a clean coding loop and makes the posterior state path auditable. [\[231\]](#) [\[217\]](#)

7. BRI credit definition

Your proposed credit formula is correct in principle: use the frequency of partial or total loss across simulations as the core loss measure. I would define it as: [\[229\]](#) [\[238\]](#) [\[239\]](#)

$$\text{Loss Frequency} = \frac{N_{\text{partial}} + N_{\text{total}}}{N_{\text{sim}}}$$

and then report:

$$\text{Containment Rate} = \frac{N_{\text{contained}}}{N_{\text{sim}}}$$

The BRI credit can then be a monotonic transformation of loss frequency, with better resilience corresponding to lower loss frequency and higher containment rate. ^[238] ^[229]

8. Minimum property fields

For v2, the minimum fire input set should be:

- PropertyType.
- OccupancyType.
- OccupancyStatus.
- HeightMeters.
- NumberOfStoreys.
- BasementPresent.
- PlantRoomLocation.
- ServiceCore.
- ConstructionType.
- WallConstruction.
- RoofType.
- FloorConstruction.
- DoorType.
- GlazingType.
- AccessType.
- EmergencyExits.
- SprinklerSystem.
- FireSuppressionType.
- SecuritySystem.
- EmergencyPower.
- FireLoadRating.
- IndustrialProcessRisk.
- ChemicalStorage.
- FireStationDistance.
- EmergencyResponseTime.
- EvacuationTime. ^[224]

9. Next explicit artefact

The next artefact should be a proper **transition matrix spec** with one row per node and one column per driver, plus suggested prior distributions for each probability. That will let your coders implement the Bayesian network without ambiguity. ^[241] ^[217] ^[231]

I can draft that next as a developer-facing table with column names, data types, priors, and formulas.

✱

So to confirm if we have a fire starting, then a high BRI score building will detect is quicker ($P(\text{Detect})$) than a low BRI score so that the fire can increase in intensity. Given intensity containment is a function of sprinkler/no sprinkler, so the intensity continues together with damage. intensity should also be a function of build materials and BRI build quality. So I think we can go to this level of detail.

Yes — that is exactly the right level of detail, and the model should be built to reflect that dynamics explicitly. In the BRD, a higher BRI score should generally mean **faster detection, better suppression, and better containment**, which reduces the time a fire can grow before control is achieved. [\[248\]](#) [\[249\]](#) [\[250\]](#)

What to encode

You should separate the fire process into three coupled quantities: $P(\text{Detect})$, $P(\text{Contain})$, and Intensity/Damage. Detection is a function of monitoring, occupancy, and building quality; containment is a function of sprinklers, compartmentation, response, and access; and intensity is a function of fuel load, construction/material resistance, and whether the fire has already escaped control. That means the BRI score should not directly set the fire outcome, but should influence these probabilities through the building's resilience features. [\[251\]](#) [\[249\]](#) [\[252\]](#) [\[253\]](#) [\[254\]](#)

Direction of effect

Your intuition is right: a better building should normally increase $P(\text{Detect})$ and $P(\text{Contain})$ while lowering the growth rate of intensity and damage accumulation. A low-BRI building should have slower detection, weaker suppression, and a higher chance that intensity continues to rise until partial or total loss occurs. That is consistent with performance-based fire engineering, which evaluates how specific building features change the fire outcome rather than relying on generic code compliance alone. [\[255\]](#) [\[256\]](#) [\[250\]](#) [\[252\]](#) [\[248\]](#) [\[251\]](#)

Practical state variables

I would express the model with four state variables evolving over time: FireIntensity, DetectionStatus, SuppressionStatus, and DamageState. At each 15-minute step, FireIntensity grows according to entry-point severity, combustibility, and material response; DetectionStatus updates through alarm/sensor/occupant pathways; SuppressionStatus updates through sprinklers and external response; and DamageState accumulates from the relationship between intensity and containment. Once suppression succeeds, intensity growth should flatten or decline; if suppression fails, the fire continues to intensify and damage continues to accumulate. [\[257\]](#) [\[258\]](#) [\[259\]](#) [\[260\]](#) [\[251\]](#)

BRI feature mapping

The BRI score should be decomposed into contributing feature groups. For example: detection quality, suppression quality, passive fire resistance, access/response quality, and continuity/operability. High-quality construction materials, better fire-resistant assemblies, better compartmentation, and better system integration should reduce the intensity growth coefficient and raise the containment probability. In other words, BRI is a latent resilience factor that modifies the fire dynamics rather than a direct outcome variable. [\[261\]](#) [\[249\]](#) [\[250\]](#) [\[252\]](#) [\[253\]](#) [\[262\]](#) [\[255\]](#) [\[248\]](#)

Suggested explicit formulation

A good implementation would look like this:

- $P(\text{Detect}_t) = f(\text{alarm}, \text{occupancy}, \text{maintenance}, \text{building_quality}, \text{BRI_detection_subscore})$.
- $P(\text{Contain}_t) = f(\text{sprinklers}, \text{suppression_type}, \text{access}, \text{response_time}, \text{compartmentation}, \text{height}, \text{BRI_protection_subscore})$.
- $\text{Intensity}_{\{t+1\}} = \text{Intensity}_t + g(\text{entry_point}, \text{fuel_load}, \text{material_resistance}, \text{suppression_effectiveness})$.
- $\text{Damage}_{\{t+1\}} = \text{Damage}_t + h(\text{Intensity}_t, \text{Containment_status}, \text{duration_uncontrolled})$.
[\[252\]](#) [\[263\]](#) [\[251\]](#)

That structure is fully compatible with your 15-minute Bayesian state progression and with a resilience credit based on simulated partial/total loss frequency. [\[264\]](#) [\[265\]](#)

Important nuance

I would avoid making BRI score directly force a lower fire intensity without explaining the mechanism. The mechanism should be material and system quality: better materials slow growth, sprinklers limit heat release, and faster detection shortens the uncontrolled phase. That makes the model defensible in a technical review, because each effect can be tied to an observable property or system attribute. [\[250\]](#) [\[262\]](#) [\[255\]](#) [\[248\]](#) [\[257\]](#)

BRD wording

A suitable BRD sentence would be: “The model shall treat BRI as a latent resilience modifier that affects fire detection probability, containment probability, and fire growth/intensity coefficients over 15-minute time steps. Fire intensity shall increase when detection and suppression fail, and shall be dampened by sprinkler presence, passive fire resistance, and building quality attributes.”. [\[249\]](#) [\[254\]](#) [\[251\]](#)

Next step

The next step should be a full **node-and-equation specification** for the Bayesian model, including how each CDM field feeds $P(\text{Detect})$, $P(\text{Contain})$, and Intensity . That is the right level to hand to the developers.

Yes agree with your summary, so lets develop a node-and-equation spec for a bayesian model

Yes — now we can specify the **node-and-equation model** at a level your team can code. The clean formulation is: BRI is a latent resilience factor that acts through detection, suppression, and material-response nodes; fire intensity is the evolving hazard state; and loss is the terminal outcome of the interaction. ^{[270] [271] [272]}

1. Core latent structure

Let R_i be the latent resilience score for property i . Decompose it into sub-scores:

$$R_i = w_D D_i + w_S S_i + w_P P_i + w_A A_i + w_C C_i$$

where D_i = detection quality, S_i = suppression quality, P_i = passive protection/material resistance, A_i = access/response quality, and C_i = continuity/recovery readiness. The weights can initially be expert-set and later calibrated to claims or incident outcomes. ^{[273] [271] [274] [275] [276]}

2. State variables

At each 15-minute step t , track:

- I_t : fire intensity.
- X_t : detected/not detected.
- U_t : internal suppression status.
- V_t : external response status.
- L_t : damage state.
- K_t : containment state. ^{[277] [278] [279]}

The model evolves as a discrete-time Bayesian dynamic system, where each node depends on the previous state and property features. ^{[280] [270]}

3. Detection equation

Use a logistic form for detection probability:

$$P(X_t = 1 \mid X_{t-1} = 0) = \sigma(\alpha_0 + \alpha_R R_i + \alpha_O O_i - \alpha_I I_t - \alpha_V Vac_i)$$

where O_i captures occupancy/monitoring quality, Vac_i captures vacancy or partial occupancy, and $\sigma(z) = 1/(1 + e^{-z})$. Higher BRI should increase detection probability through better alarms, monitoring, and management quality. ^{[271] [272] [275] [276] [273]}

4. Intensity growth equation

Define intensity growth as:

$$I_{t+1} = I_t + \exp(\beta_0 + \beta_E E_i + \beta_F F_i - \beta_P P_i - \beta_U U_t - \beta_V V_t)$$

where E_i is ignition/exposure severity, F_i is fuel/load potential, P_i is passive resistance/material quality, and U_t, V_t are internal and external suppression states. This makes intensity grow faster

when materials are weak and slower when protection is strong. ^[281] ^[282] ^[283] ^[284] ^[285] ^[286]

5. Internal suppression equation

Model sprinkler or internal suppression success as:

$$P(U_t = 1 \mid X_t = 1) = \sigma(\gamma_0 + \gamma_S Spr_i + \gamma_T T_{resp} + \gamma_C Comp_i + \gamma_R R_i)$$

where Spr_i is sprinkler availability/effectiveness, T_{resp} is internal response delay, and $Comp_i$ is compartmentation quality. If sprinklers are present and effective, the probability of control rises materially. ^[287] ^[288] ^[286] ^[289]

6. External response equation

Model fire service success as:

$$P(V_t = 1 \mid X_t = 1) = \sigma(\delta_0 + \delta_A Acc_i + \delta_H H_i - \delta_E ERT_i - \delta_F Fdist_i)$$

where Acc_i is access quality, H_i is height effect, ERT_i is emergency response time, and $Fdist_i$ is fire station distance. For high-floor incidents, include a vertical penalty term: ^[290] ^[291] ^[292]

$$H_i = \log(1 + \text{NumberOfStoreys}) + \phi \cdot \text{FloorLevelMeters}$$

so upper floors are harder to control externally. ^[292] ^[293] ^[290]

7. Containment equation

Containment should depend on successful suppression and the current intensity:

$$P(K_t = 1) = \sigma(\kappa_0 + \kappa_U U_t + \kappa_V V_t + \kappa_P P_i - \kappa_I I_t)$$

If this probability is low and intensity remains above threshold, the event transitions toward partial or total loss. This is the core BRI threshold logic. ^[278] ^[279] ^[294] ^[295] ^[271]

8. Damage accumulation equation

Damage should accumulate monotonically:

$$L_{t+1} = L_t + \lambda_0 + \lambda_I I_t - \lambda_K K_t$$

with a floor at 0 and a cap at total loss. The higher the uncontrolled intensity and the lower the containment, the faster damage rises. You can classify `PartialLoss` once L_t exceeds a lower bound, and `TotalLoss` once it exceeds the terminal threshold. ^[296] ^[279] ^[297] ^[281] ^[270]

9. State transition summary

At each time step:

1. Update detection probability.
2. If detected, update internal suppression probability.
3. In parallel or subsequently, update external response probability.
4. Recompute fire intensity using material and suppression factors.

5. Recompute containment probability.

6. Update damage.

7. Classify outcome. ^[279] ^[273] ^[278]

That sequence is directly implementable and matches the logic you described.

10. Property fields to drive the equations

For v2, the relevant CDM fields map to the equations as follows:

- **Detection:** SecuritySystem, OccupancyStatus, OperatingHours, BusinessContinuity, EmergencyPower.
- **Suppression:** SprinklerSystem, FireSuppressionType, ConstructionType, WallConstruction, FloorType, RoofType, LastMajorWorksDate.
- **Passive resistance:** ConstructionType, PropertyPeriod, WallConstruction, RoofType, GlazingType, DoorType, FloorConstruction.
- **External response:** AccessType, HeightMeters, NumberOfStoreys, FloorLevelMeters, FireStationDistance, EmergencyResponseTime.
- **Ignition/exposure:** BasementPresent, PlantRoomLocation, FireLoadRating, IndustrialProcessRisk, ChemicalStorage, OccupancyType. ^[298]

11. Output measures

The model should return:

- $P(\text{Contained})$.
- $P(\text{PartialLoss})$.
- $P(\text{TotalLoss})$.
- Mean time-to-containment.
- Mean time-to-overwhelm.
- Loss-frequency-based BRI fire credit. ^[297] ^[299] ^[300]

12. Recommended coding unit

The best first coding unit is a `FireScenario` object containing the property features, current state, and transition equations, plus a `run()` method that advances the scenario in 15-minute steps. The second unit is a `FirePortfolioModel` that runs many scenarios and aggregates the loss frequencies into the final BRI score. ^[276] ^[270] ^[273]

I can next turn this into a **developer-ready schema**: node list, equation list, and JSON-like field definitions for the v2 workbook.

remember to separate out the modelling into component models, a small number no more than 3 or 4. Models can then go through a separate model governance process.

Absolutely — that is the right governance design.

The best structure is to split the overall fire engine into 3 or 4 **component models**, each with a clear purpose, inputs, outputs, and independent validation path. That makes the system easier to govern, calibrate, and defend in review.

Recommended split

1. Hazard initiation model.

This estimates how a fire starts or enters the building, by pathway such as internal ignition, external exposure, basement/service-space ignition, or upper-floor ignition.

2. Fire progression model.

This simulates how intensity evolves over time, in 15-minute steps, based on fuel load, materials, compartmentation, and whether suppression is working.

3. Detection and response model.

This estimates the probability and timing of detection, internal suppression, and external fire service intervention, including the penalty for height and access constraints.

4. Loss / threshold model.

This converts the progression path into partial loss or total loss, using the BRI exceedance threshold and time-to-containment logic.

Why this is better

This decomposition keeps each model small enough to govern properly. Each component can have its own priors, calibration data, assumptions, and versioning, instead of trying to approve one large monolithic model.

It also matches the way the problem actually works:

- initiation determines the starting condition,
- progression determines how fast the event grows,
- response determines whether control is achieved,
- loss determines the terminal outcome.

Governance benefit

Each component should be treated as a separately governed model artefact with:

- a model purpose statement,
- input/output specification,
- calibration method,

- validation tests,
- limits of use,
- approval version.

Then the overall BRI fire score becomes an orchestration layer that combines the outputs of the component models rather than a single opaque engine.

Practical recommendation

For the BRD, I would define the architecture as:

- **Model A: Fire Initiation**
- **Model B: Fire Growth and Progression**
- **Model C: Detection, Suppression and External Response**
- **Model D: Loss Threshold and Resilience Credit**

If you want to keep it to only three, then combine A and B into one **Hazard Model**, and keep C and D separate.

Suggested wording for the BRD

“The fire assessment capability shall be implemented as a modular architecture consisting of no more than four separately governed component models. Each component model shall be independently specified, calibrated, validated, version-controlled, and approved. The overall resilience credit shall be generated by orchestrating the outputs of the component models.”

That is the cleanest way to make the system defensible to technical review, model risk governance, and future audit.

Interesting. The hazard initiation model starts with 10,000 events where a subset will kickoff a fire from external or internal. This allows the fire hazard to be pari-passu with storm and hence aggregatable.

Yes — that is a sound architecture, and it makes the fire module structurally parallel to storms. You would run 10,000 scenarios, let only a subset initiate an internal or external fire, and then treat the initiation layer as the analog of storm-event generation so the hazard can be aggregated consistently across BRI modules. ^[306] ^[307] ^[308]

Hazard initiation layer

The initiation model should be a frequency-and-classification layer. Each simulation draw first decides whether the event is a fire at all, and if so whether it is internal, external, concealed, or process-related, using priors based on the property and exposure context. That makes the fire hazard pari-passu with your storm hazard logic because both start as large Monte Carlo ensembles with only some scenarios becoming loss-producing events. ^[307] ^[308] ^[309] ^[306]

Why this works

Fire risk assessment literature already separates initiating-event frequency from consequence modeling, which is exactly the split you need here. In one widely used framework, risk is the product of scenario frequency and scenario consequence, and event-tree logic is used to enumerate the possible fire pathways after ignition. Your fire initiation model therefore sits upstream of the Bayesian progression model, not inside it. [\[306\]](#) [\[307\]](#)

Parallel with storms

This gives you the same architecture as your storm engine: a top-level stochastic event generator, then a conditional response model, then a severity/loss model. For storms, the initiating variables are precipitation and wind; for fire, the initiating variables are ignition likelihood, exposure route, and scenario class. Once you do that, the outputs can be aggregated into a common BRI resilience framework across hazards. [\[308\]](#) [\[309\]](#) [\[310\]](#) [\[311\]](#)

Practical model split

I would formalize the fire side as:

- **Model A: Initiation / Exposure Frequency.**
- **Model B: Fire Progression and Intensity.**
- **Model C: Detection, Suppression, and External Response.**
- **Model D: Containment, Loss, and Resilience Credit.**

Model A is the point where only a subset of the 10,000 Monte Carlo draws become actual fire scenarios, exactly as you described. [\[307\]](#) [\[306\]](#)

BRD wording

A suitable BRD statement would be: “The fire hazard module shall generate a large ensemble of stochastic scenarios, of which only a calibrated subset shall instantiate fire initiation events. The model shall then condition fire progression, response, and loss on the initiating scenario type, enabling direct aggregation with other BRI hazard modules.” [\[310\]](#) [\[306\]](#) [\[307\]](#)

Next step

The next thing to define is the **initiation event taxonomy and priors**: the probability of internal vs external initiation, and the property fields that move those priors up or down. That will let your team code the first model cleanly.

✱

Defining the probability distribution for fire ignition event frequency

A practical way to define this is to model fire ignition frequency as a frequency distribution over scenario classes, not as a single scalar. For BRI, the most useful formulation is a Poisson or mixed-Poisson event process for “fire initiation events,” with scenario-specific probabilities for internal, external, and concealed ignition pathways. ^[323] ^[324] ^[325]

Recommended distribution

Use a two-stage distribution:

1. Annual or portfolio event frequency λ_i for property i .
2. Conditional initiation class probabilities $P(c \mid \lambda_i)$, where c is internal, external, basement/service, electrical, process-related, or wildfire exposure. ^[324] ^[325]

A simple implementation is:

$$N_i \sim \text{Poisson}(\lambda_i)$$

where N_i is the number of ignition events over the simulation horizon, and each event is then assigned a type using a categorical distribution. That gives you the event-frequency layer you need to keep fire pari-passu with storm hazard sampling. ^[326] ^[327] ^[323] ^[324]

What drives λ_i

The ignition frequency should be a function of exposure, occupancy, process risk, maintenance, and building condition. For commercial property, the relevant drivers are things like OccupancyType, OccupancyStatus, IndustrialProcessRisk, ChemicalStorage, OperatingHours, FireLoadRating, BasementPresent, PlantRoomLocation, ConstructionType, and ClaimsHistory. ^[328] ^[329] ^[330]

A defensible parametric form is:

$$\log(\lambda_i) = \beta_0 + \beta^\top x_i$$

where x_i is the property feature vector. ^[329] ^[331]

Conditional initiation classes

Once an ignition event is sampled, classify it into one of the following:

- InternalElectrical.
- Kitchen/occupant.
- PlantRoom/technical.
- Process/industrial.
- ExternalExposure.
- Basement/service-space.
- WUI/external ember or adjacent fire. ^[327] ^[332] ^[324]

Then assign:

$$P(c \mid x_i) = \text{softmax}(Wx_i)$$

or use a simpler rule-based categorical table for v1. This makes the model explicit about how the fire started, which matters for downstream detection and containment. ^[333] ^[334] ^[323] ^[324]

Suggested priors

For an initial BRI implementation, set priors by property class rather than trying to estimate everything from scratch. For example, high-process or high-load assets should have higher priors for internal ignition, while assets with high external exposure or poor adjacency control should have higher priors for external initiation. The distribution can then be updated using claims history and incident evidence. ^[335] ^[325] ^[328] ^[329]

Practical BRD wording

You can specify it as:

“The fire initiation model shall simulate ignition events as a stochastic frequency process. For each asset, the model shall sample the number of initiating events from a calibrated frequency distribution and assign each event to an initiation class conditional on property attributes. The initiation distribution shall be consistent with the hazard-module architecture and shall support aggregation with other BRI hazard classes.”. ^[325] ^[324] ^[326]

Best default choice

If you want one concrete starting point, use:

- Poisson for the count of ignition events.
- Categorical for initiation class.
- Logistic or softmax links for property-driven class probabilities. ^[331] ^[329] ^[324]

That is simple, governable, and easy to calibrate later.

Next step

The next thing to define is the **actual prior table** for each initiation class, by property type and by CDM field. That is where we turn the distribution into a code-ready specification.

**

yes poisson is absolutely the correct distribution choice!

Yes — and the literature supports using Poisson for ignition frequency as the core event model for building fires. That is the right default for BRI because fire starts are rare, countable events over time, and the model can then condition on scenario class and property attributes. ^[340] ^[341] ^[342]

Recommended ignition model

For each property i , define an annual ignition frequency λ_i . Then sample ignition counts as:

$$N_i \sim \text{Poisson}(\lambda_i)$$

with λ_i estimated from the property's features, history, and exposure context. This is consistent with fire risk analysis practice, where initiating-event frequency is treated separately from consequences. [\[343\]](#) [\[344\]](#) [\[345\]](#) [\[340\]](#)

What λ_i should represent

λ_i should be the expected number of fire ignition events per year for that property, before any response or mitigation effects are applied. It should reflect internal fire triggers, external exposure, and process-related ignition sources, but not sprinkler performance or containment success, which belong in downstream models. [\[342\]](#) [\[346\]](#) [\[347\]](#)

Suggested parameterization

A practical form is:

$$\log(\lambda_i) = \beta_0 + \beta^\top x_i$$

where x_i includes property-level features such as occupancy, process risk, chemical storage, operating hours, construction type, fire load, and claim history. This lets high-risk properties carry higher fire frequencies while still using the same Poisson backbone across the portfolio. [\[341\]](#) [\[348\]](#)
[\[344\]](#) [\[349\]](#) [\[350\]](#)

Initiation classes

After an ignition event is drawn, classify it into a pathway such as internal electrical, kitchen/occupant, plant-room, process-related, basement/service-space, or external exposure. You can model the class as categorical conditional on the property features, which preserves the fire origin information needed for detection and response modelling. [\[351\]](#) [\[345\]](#) [\[352\]](#) [\[342\]](#)

BRI-specific use

This is exactly the right structure if you want fire to be pari-passu with storms: a top-level event frequency model followed by conditional severity and loss models. The output then aggregates cleanly across hazards because each hazard module begins with a stochastic event count and proceeds to outcome states. [\[353\]](#) [\[354\]](#) [\[355\]](#) [\[356\]](#)

BRD wording

A clean BRD statement would be: “The fire initiation module shall model ignition frequency as a Poisson process with property-specific intensity. The model shall estimate annual ignition frequency from building attributes and historical evidence, then assign each ignition to a scenario class for downstream progression and loss modelling.”. [\[340\]](#) [\[343\]](#) [\[342\]](#)

Next step

The next thing to define is the **actual prior table** for λ_i by property class, plus the modifiers that raise or reduce ignition frequency. That will make the initiation model fully codeable.

✱

Yes lets have a look at the actual prior table.

Yes — and for BRI, the right thing is not to invent a frequency prior from scratch, but to start with a **baseline prior table** by asset class and then let property features scale it up or down. The fire-practice literature treats ignition frequency as a calibrated prior over bins or source classes, with Bayesian updates from operating experience and sparse-data handling where needed. [365] [366]

How to structure the prior

Use a prior on annual ignition frequency λ for each building class, then apply multiplicative modifiers for property condition, occupancy, process load, and protection quality. In the simplest form: [367] [368]

$$\lambda_i = \lambda_{0,c} \times m_{proc} \times m_{occ} \times m_{maint} \times m_{exposure} \times m_{history}$$

where $\lambda_{0,c}$ is the class prior for property class c . That mirrors the Bayesian-update logic used in fire PRA frequency estimation. [366] [369] [365]

Suggested prior table

Below is a practical **starting prior table** for commercial property classes. These are not “truth values”; they are deliberately usable seed priors to be refined with your own experience data.

Property class	Baseline annual ignition frequency $\lambda_{0,c}$	Typical interpretation
Retail unit	0.03	Low to moderate ignition likelihood; occupant/plug-load driven [370] [369]
Office block	0.02	Lower than retail if process risk is minimal [370] [369]
Industrial unit	0.06	Higher due to equipment, maintenance, and process loads [370] [371]
Warehouse	0.05	Higher due to load, storage, hot work, and access issues [370] [372]
Mixed use	0.04	Blended prior; depends on dominant use [370]
Leisure	0.03	Occupancy-related and kitchen/electrical sources [370]
Healthcare	0.025	Lower ignition frequency but potentially higher consequence [370]

These priors are compatible with a Poisson annual count model and can be adjusted later by class, country, occupancy, and evidence history. [366] [367]

Modifier table

A more useful next layer is the multiplicative modifier table.

Driver	Example modifier	Effect
High industrial process risk	1.5 to 3.0	Raises ignition frequency materially [370] [371]
Chemical storage present	1.2 to 2.0	Increases ignition likelihood [370] [373]
High occupancy / extended hours	1.1 to 1.6	More ignition opportunity [370] [369]
Poor maintenance / high claims history	1.2 to 2.5	Raises prior frequency [370] [366]
Strong fire separation / protected plant	0.7 to 0.9	Reduces effective ignition frequency [370]
Good housekeeping / low transient load	0.6 to 0.9	Reduces transient ignition opportunity [374] [365]
Vacant or partially vacant	0.8 to 1.2	Depends on security and arson exposure [370]

If you want the model to be governable, keep modifiers coarse at first and avoid too many finely tuned subcategories. [\[365\]](#) [\[366\]](#)

Class-specific initiation priors

You should also define the probability of each ignition class conditional on building type.

Initiation class	Retail	Office	Industrial	Warehouse	Mixed
Internal electrical	0.30	0.35	0.25	0.20	0.28
Kitchen / occupant	0.30	0.10	0.10	0.05	0.12
Plant / technical room	0.10	0.15	0.20	0.20	0.15
Process-related	0.05	0.05	0.25	0.25	0.12
Basement / service space	0.10	0.15	0.10	0.10	0.10
External exposure	0.10	0.10	0.05	0.15	0.08
Other / unknown	0.05	0.10	0.05	0.05	0.15

These are starter priors only, but they make the model operational immediately and allow later calibration with claims or incident records. [\[375\]](#) [\[366\]](#)

Bayesian update rule

The property-specific prior can be updated as:

$$\log(\lambda_i) = \log(\lambda_{0,c}) + \sum_j \beta_j x_{ij}$$

where each x_{ij} is a feature flag or score from the CDM. A claim history term and a maintenance-quality term should have the strongest influence because they are usually the best available empirical proxies for observed incident propensity. [\[368\]](#) [\[369\]](#) [\[366\]](#)

BRD wording

You can write this into the BRD as:

“The fire initiation model shall use class-based Poisson priors for annual ignition frequency, calibrated by property class and adjusted through multiplicative Bayesian modifiers derived from property attributes, usage, maintenance, process load, and historical incident experience.”^[367] ^[365]
^[366]

What to do next

The next step is to turn this into a **formal prior matrix** for the CDM, with one row per field and a coefficient or modifier range for each. That will let your team implement the initiation model directly.

✱

Here is the updated CDM, a few more changes than previously suggested. Next stage is the implementing modifier logic or occupancy and process load

Yes — v2 gives us the right basis to implement the **occupancy and process-load modifiers** as explicit priors, and it also makes the model governance cleaner because the fire module can now use the fire-specific BRI fields directly. The key change is that v2 separates the generalized commercial attributes from the resilience and hazard profile, so we can drive ignition frequency and progression from a smaller, better-defined set of nodes.^[383]

What changed in v2

The most important fire-relevant additions in v2 are CommercialType, OccupancyStatus, NumberOfStoreys, PropertyCondition, LastMajorWorksDate, PropertyPeriod, AutomaticDetectionInstalled, SuppressionSystemsInstalled, StructuralFireResistanceAdequate, CompartmentsProvided, FireStoppingAtPenetrations, ExternalMaterialsFireResistant, FireHazardClass, BRIFireScore, and the fire-related FireCodes field. Several v1 fields that were directly useful for fire — such as OccupancyType, OperatingHours, FireLoadRating, FireStationDistance, SprinklerSystem, FireSuppressionType, SecuritySystem, and EmergencyPower — no longer appear as standalone fields in the v2 summary you shared, so we should rebuild those concepts from the new structure rather than assume the old names still exist.^[383]

Occupancy modifier

For the initiation model, occupancy should be implemented as a **frequency multiplier** on the Poisson ignition rate, not as a direct loss variable. A sensible first formulation is:

$$\lambda_i = \lambda_{0,c} \times m_{occ} \times m_{proc} \times m_{cond} \times m_{protection} \times m_{history}$$

where CommercialType, OccupancyStatus, and PropertyCondition drive m_{occ} and m_{cond} . The idea is that more active, more complex, or more transiently occupied buildings should have higher

ignition opportunity, while vacant or low-activity assets should generally have lower ignition rates unless security/arson risk or poor maintenance pushes the opposite way. ^[384] ^[385] ^[386] ^[383]

A practical occupancy table can start like this:

CommercialType / status	Modifier	Interpretation
Office, fully occupied	1.00	Baseline active commercial occupancy
Office, partially vacant	0.90	Fewer ignition opportunities but maintenance uncertainty remains
Office, vacant	0.80	Lower internal ignition opportunity, but longer unnoticed events possible
Retail, fully occupied	1.15	More occupant activity and plug-load diversity
Retail, partially vacant	1.05	Mixed occupancy exposure
Retail, vacant	0.85	Lower ignition opportunity, higher delay-to-detection risk
Hotel / leisure, fully occupied	1.20	Higher occupancy-driven ignition opportunity
Healthcare, fully occupied	1.10	More systems and high operational complexity
Industrial / mixed process	1.40 to 2.50	Much higher ignition opportunity depending on process load
Warehouse / storage	1.20 to 1.80	Load and hot-work risk, often low supervision periods

These are seed values, not final calibrated priors, but they are appropriate for v2 implementation and later adjustment. ^[387] ^[388]

Process-load modifier

Because v2 removed the old FireLoadRating and IndustrialProcessRisk fields, the process-load concept needs to be inferred from a combination of CommercialType, UseClassUK0, BusinessRatesCategory, AnchorTenant, VacancyCount, TenancyDuration, and potentially the new FireCodes/CommercialType grouping. For a first version, I would create a latent ProcessLoadScore derived from those inputs, then map it to a multiplicative modifier: ^[383]

$$m_{proc} = \exp(\beta_{proc} \cdot \text{ProcessLoadScore})$$

with ProcessLoadScore on a 0–1 or 0–5 scale. In practice, the score should rise for industrial, storage, restaurant, leisure-with-kitchen, and high-occupancy mixed-use assets, and be lower for low-load office assets. ^[389] ^[390] ^[391] ^[383]

A simple process-load table could be:

Derived process class	Modifier
Low-load office / professional	1.00
Light retail / standard office	1.10
Restaurant / leisure / food service	1.25
Storage / warehouse	1.35
Light industrial	1.50
Heavy industrial / process-intensive	1.75 to 2.50

Property condition modifier

PropertyCondition should affect both initiation and progression. On initiation, poorer condition should increase ignition frequency because of maintenance defects, degraded systems, and housekeeping uncertainty. A first-pass modifier could be: [\[387\]](#) [\[383\]](#)

PropertyCondition	Modifier
Excellent	0.85
Good	0.95
Fair	1.05
Poor	1.20
Very poor	1.40

If you prefer, this can be decomposed into separate modifiers for ignition and progression, with the initiation side being smaller and the progression side larger. [\[392\]](#) [\[393\]](#)

Protection modifier

The new v2 resilience fields are more important than before because they let us convert building quality into explicit protective effects. For fire, the most relevant are AutomaticDetectionInstalled, SuppressionSystemsInstalled, StructuralFireResistanceAdequate, CompartmentsProvided, FireStoppingAtPenetrations, and ExternalMaterialsFireResistant. These should not reduce ignition frequency directly in every case, but they should reduce the probability that an ignition becomes a damaging event and should materially reduce growth intensity once a fire begins. [\[394\]](#) [\[395\]](#) [\[396\]](#) [\[383\]](#)

For the initiation prior, I would use a modest protection modifier if the intent is to capture better housekeeping and managed risk, for example:

Protection level	Initiation modifier
Not assessed / absent	1.10
Partial	1.00
Meets minimum	0.95
Enhanced	0.90
Verified	0.85

Then use a stronger protection effect in the progression model rather than trying to force all resilience into the ignition model. [\[397\]](#) [\[394\]](#)

History modifier

Your historical fields remain useful: LastFireDate, FireDamageSeverity, and ClaimsHistory are ideal priors for Bayesian updating. A straightforward history factor is: [\[383\]](#)

$$m_{history} = \exp(\beta_h \cdot H_i)$$

where H_i is a score built from recency and severity of prior fire claims or incidents. A property with a recent severe claim should move up materially; a long clean history can reduce the prior slightly, but I would keep that effect modest unless you have strong calibration evidence. ^[386] ^[388] ^[390] ^[387]

Recommended implementation

For v2, I would implement the fire initiation prior as:

$$\lambda_i = \lambda_{0,c} \times m_{occ} \times m_{proc} \times m_{cond} \times m_{protection} \times m_{history}$$

where:

- $\lambda_{0,c}$ comes from the commercial asset class prior table.
- m_{occ} comes from CommercialType and OccupancyStatus.
- m_{proc} comes from a derived process-load score.
- m_{cond} comes from PropertyCondition and maybe LastMajorWorksDate.
- $m_{protection}$ comes from fire-protection fields.
- $m_{history}$ comes from incident and claims history. ^[384] ^[387] ^[383]

That gives you a clean and governable layer before you move into the 15-minute Bayesian progression model.

Next artifact

The next thing to produce is the actual **modifier matrix** for v2: one row per CDM field, with a suggested direction, scale, and whether it affects initiation, progression, detection, or containment. That will let the developers wire the model directly from the spreadsheet.

✱

OK, onto the modifier matrix

Yes — the modifier matrix should now be explicit, field-by-field, and split by **model component** so governance stays manageable. The right structure is to assign each CDM feature to one of four functions: initiation, progression, detection/response, or containment/loss. ^[398] ^[399] ^[400]

Modifier matrix

CDM field	Model component	Direction	Suggested first-pass effect
CommercialType	Initiation	Up/down	Sets baseline ignition prior by class ^[401]
OccupancyStatus	Initiation	Down if vacant, up if fully occupied	Affects ignition opportunity and detection delay ^[401] ^[402]

CDM field	Model component	Direction	Suggested first-pass effect
PropertyCondition	Initiation / progression	Worse = up	Poorer condition raises ignition and spread risk [401] [403]
LastMajorWorksDate	Progression / response	Older = up	Older fabric and systems are more failure-prone [401]
NumberOfStoreys	Response / containment	Up	More floors increase vertical spread and response difficulty [401] [402]
BasementPresent	Initiation / progression	Up	Basements increase concealed spread and delayed discovery risk [401]
PlantRoomLocation	Initiation / progression	Basement/roof = up	Technical-space fires often develop faster or harder to isolate [401]
ServiceCore	Progression / containment	Central core can be neutral/mixed	Core configuration affects compartmentation and smoke spread [401]
ConstructionType	Progression	Stronger/fire-resistive = down	Better construction slows fire growth and structural compromise [401] [404]
FloorType	Progression	Stronger = down	Ground floor system influences spread and collapse resilience [401]
FloorConstruction	Progression	Stronger = down	More fire-resistive floors reduce escalation [401]
RoofDetails fields	Progression	More combustible = up	Roof assemblies are major drivers of spread [401] [405]
SoftStory	Progression / loss	True = up	Structural vulnerability raises total-loss probability [401]
ShapeIrregularity	Progression / loss	Greater irregularity = up	Irregular form can worsen fire and post-fire stability [401]
BrickVeneer	Progression	True = mixed/up	Cladding can affect fire spread and detachment risk [401]
GlassType	Progression / containment	Better resistance = down	Better glazing reduces external flame/smoke entry [401]
RetrofitYear	Progression / response	Recent = down	Recent retrofit should improve resistance and systems [401]
HasCrippleWall	Loss	True = up	Adds structural fragility in multi-hazard contexts [401]

CDM field	Model component	Direction	Suggested first-pass effect
Accessible features	Response	Better = down	More exits/lifts improve response and evacuation [401]
CommercialType+UseClassUKO+BusinessRatesCategory	Initiation	Up/down	Drives latent process-load score [401]
AnchorTenant	Initiation / condition	Stable tenant can lower up	Stable tenancy usually lowers vacancy-related risk [401]
WAULT	Initiation / condition	Longer = down	Longer leases tend to reduce vacancy churn risk [401]
VacancyCount	Initiation / detection	Higher = up	Vacancy increases ignition uncertainty and delayed detection [401]
ClaimsHistory	All components	Higher = up	Prior claims are strong Bayesian evidence of elevated risk [401]
FireHazardClass	Prior setting	Higher class = up	Sets hazard prior for the initiation model [401]
BRIFireScore	All components	Higher score = down	Use as latent resilience modifier, not direct output [401]
AutomaticDetectionInstalled	Detection	Present = down	Faster detection and alarm activation [401]
SuppressionSystemsInstalled	Response / containment	Present = down	Reduces intensity growth and speeds control [401]
StructuralFireResistanceAdequate	Progression / loss	Present = down	Slows structural failure and total-loss transition [401]
CompartmentsProvided	Progression / containment	Present = down	Limits spread between zones/floors [401] [405]
FireStoppingAtPenetrations	Progression / containment	Present = down	Prevents hidden spread through services [401]
ExternalMaterialsFireResistant	Progression	Present = down	Reduces façade/roof contribution to spread [401] [405]
AccessRouteResilient	External response	Present = down	Improves fire brigade access and intervention [401] [406]
BusinessContinuityPlanInPlace	Loss / recovery	Present = down	Doesn't stop fire, but reduces effective business-loss severity [401]
EmergencyProceduresTested	Detection / response	Present = down	Better practiced response reduces uncontrolled time [401]

How to use it

For v2, I recommend each row be mapped to one of three object types: **prior modifier**, **transition modifier**, or **loss modifier**. That keeps the model clean and avoids mixing initiation with response or structural vulnerability. ^[400] ^[398]

Suggested factor structure

A practical factor decomposition is:

- InitiationFactor = occupancy + process + vacancy + history.
- DetectionFactor = automatic detection + procedures + monitoring.
- ResponseFactor = suppression + access + height + brigade reach.
- ProgressionFactor = materials + compartments + structural fire resistance + roof/facade.
- LossFactor = structural fragility + continuity + recovery readiness.

That structure is consistent with performance-based fire engineering, where fire safety is achieved by combining active and passive measures to control ignition, spread, suppression, evacuation, and recovery. ^[407] ^[404] ^[405]

Suggested coding rule

Each CDM field should contribute either:

1. a direct multiplier,
2. a categorical prior shift, or
3. a binary transition boost/reduction.

For example, `SuppressionSystemsInstalled=True` should reduce the growth rate and increase containment probability, while `OccupancyStatus=Vacant` should reduce ignition probability but may increase detection delay if paired with weak monitoring. ^[402] ^[401]

Governance rule

Do not let one field affect too many components unless there is a clear physical reason. For instance, `PropertyCondition` can reasonably affect both initiation and progression, but `BusinessContinuityPlanInPlace` should mainly affect loss/recovery rather than ignition. That will make the model easier to validate and explain in governance review. ^[404] ^[398]

Next step

The next deliverable should be the **numeric modifier ranges** for each field, grouped by component model. Then we can convert this matrix into actual equations and defaults for the developers.

Drafting numeric scaling ranges for initiation risk factors

Yes — for the initiation layer, the cleanest approach is to define **numeric multipliers on the Poisson baseline** rather than trying to directly score the risk on an arbitrary 0–100 scale. That keeps the model physically interpretable and makes governance easier because each factor says how much it increases or decreases the annual ignition rate. ^[414] ^[415]

Scaling principle

Use multiplicative factors centered around 1.0:

- values below 1.0 reduce ignition frequency,
- values above 1.0 increase ignition frequency,
- the overall frequency is the product of all selected modifiers times the class prior. ^[416] ^[417]

That is consistent with fire-practice frequency modeling, where the starting point is an annual ignition frequency and then condition-specific evidence shifts that frequency upward or downward. ^[418] ^[419]

Recommended numeric ranges

Below is a workable first-pass range set for the initiation model.

Factor	Range	Interpretation
Occupancy activity	0.75 to 1.30	Lower for vacant/low-activity assets, higher for active, dense, or high-churn occupancy
Process load	1.00 to 2.50	Higher for restaurant, industrial, storage, and process-intensive uses
Property condition	0.85 to 1.40	Worse condition raises ignition likelihood
Vacancy / churn	0.80 to 1.25	Vacancy reduces some ignition opportunities but can raise undetected-fire and malicious-event risk
Housekeeping / management proxy	0.85 to 1.20	Better management reduces ignition chance
Historical claims / incidents	0.90 to 1.60	Recent or severe fire history raises prior
Protection quality	0.85 to 1.05	Slight effect on initiation; stronger effects should be reserved for progression and containment

These ranges are intentionally moderate. The initiation layer should not over-explain the model; its job is to set the event frequency, while progression and response should carry most of the resilience effect. ^[420] ^[421]

Occupancy scaling

For OccupancyStatus, I would start with:

OccupancyStatus	Multiplier
Fully occupied	1.00
Partially vacant	0.90
Vacant	0.80 to 0.95

The lower bound should only be used if the property has strong monitoring and security; otherwise vacancy can increase loss severity even if ignition frequency is lower. [\[422\]](#) [\[423\]](#)

Process-load scaling

For the process load layer, use a stronger range:

Process class	Multiplier
Low-load office	1.00
Standard retail / office	1.10
Hotel / leisure / food service	1.20 to 1.35
Warehouse / storage	1.25 to 1.50
Light industrial	1.50 to 1.80
Heavy industrial / process	1.80 to 2.50

This is the most important initiation driver after base property type because it captures combustible inventory, equipment, hot works, and operational complexity. [\[424\]](#) [\[425\]](#) [\[426\]](#)

Condition and history scaling

For PropertyCondition:

Condition	Multiplier
Excellent	0.85
Good	0.95
Fair	1.05
Poor	1.20
Very poor	1.40

For ClaimsHistory / LastFireDate:

History state	Multiplier
No prior incidents	1.00
Old minor incident	1.05

History state	Multiplier
Recent minor incident	1.10
Recent moderate/severe incident	1.25 to 1.60

This fits the Bayesian view that recent evidence should pull the prior materially more than distant evidence. [\[417\]](#) [\[418\]](#)

Recommended composite formula

The initiation frequency should be:

$$\lambda_i = \lambda_{0,c} \times m_{occ} \times m_{proc} \times m_{cond} \times m_{hist} \times m_{prot}$$

where $\lambda_{0,c}$ is the asset-class prior and the multipliers are constrained to the ranges above. This is simple enough for coding, but still flexible enough to calibrate later. [\[415\]](#) [\[419\]](#) [\[414\]](#) [\[418\]](#)

Governance guidance

Keep these initiation multipliers **narrower** than the progression and response multipliers. If you let occupancy or condition drive the whole model too aggressively, you will blur the boundary between ignition frequency and fire control, which makes validation harder. The initiation layer should tell you how often fires begin; the response model should tell you whether they become damaging. [\[421\]](#) [\[420\]](#)

Next step

The next artifact should be the **numeric scaling ranges for progression and response** — especially for AutomaticDetectionInstalled, SuppressionSystemsInstalled, CompartmentsProvided, StructuralFireResistanceAdequate, FireStoppingAtPenetrations, and AccessRouteResilient. That is where the biggest resilience effects will sit.

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yes numeric scaling ranges for progression and response

Yes — for progression and response, the numeric ranges should be much stronger than the initiation layer because this is where BRI actually changes the outcome. The point is to make sprinklers, compartmentation, fire resistance, and access produce large enough shifts in containment probability and intensity growth that the model meaningfully distinguishes AA from lower-rated assets. [\[435\]](#) [\[436\]](#) [\[437\]](#)

Recommended scaling ranges

Use multipliers centered on 1.0, but allow a wider range than for initiation.

Factor	Range	Interpretation
Automatic detection	0.70 to 0.95	Earlier warning shortens uncontrolled growth [438] [439]

Factor	Range	Interpretation
Suppression systems	0.50 to 0.90	Sprinklers/active suppression materially reduce escalation [438] [440]
Structural fire resistance	0.60 to 0.95	Better resistance slows collapse and major damage [438] [441]
Compartments provided	0.55 to 0.90	Compartmentation limits spread between zones/floors [438] [442]
Fire stopping at penetrations	0.70 to 0.95	Reduces concealed spread [438] [442]
External materials fire resistant	0.65 to 0.95	Reduces façade/roof contribution to escalation [438] [442]
Access route resilient	0.70 to 0.95	Improves fire brigade intervention success [438] [443]
Emergency procedures tested	0.80 to 0.97	Better drills shorten response time [438] [439]
Business continuity plan	0.85 to 0.98	Mainly reduces effective loss severity, not ignition [438]
Height / upper-floor penalty	1.05 to 1.60	Higher floors worsen external response and control [438] [444]

Detection scaling

AutomaticDetectionInstalled should mostly affect the detection-time distribution rather than the ignition rate. A practical multiplier is:

Condition	Detection multiplier
Not assessed / absent	1.00
Partial	0.90
Meets minimum	0.80
Enhanced	0.72
Verified	0.65

This means a verified detection system should make fire discovery materially faster, which is consistent with Bayesian fire response models and performance-based design thinking. [\[445\]](#) [\[439\]](#)

Suppression scaling

SuppressionSystemsInstalled should have the largest single effect on progression. A workable multiplier is:

Condition	Suppression multiplier
Not assessed / absent	1.00
Partial	0.85
Meets minimum	0.75
Enhanced	0.62
Verified	0.50

This is where sprinklers need to bite, because the literature consistently treats suppression as the main brake on fire escalation. [\[446\]](#) [\[447\]](#) [\[440\]](#)

Containment / compartmentation scaling

For CompartmentsProvided:

Condition	Containment multiplier
Not assessed / absent	1.00
Partial	0.88
Meets minimum	0.75
Enhanced	0.65
Verified	0.55

For FireStoppingAtPenetrations:

Condition	Multiplier
Not assessed / absent	1.00
Partial	0.92
Meets minimum	0.82
Enhanced	0.74
Verified	0.68

This is particularly important because concealed spread through services is a classic failure mode in compartmented buildings. [\[437\]](#) [\[442\]](#)

Structural resistance scaling

For StructuralFireResistanceAdequate:

Condition	Structural-loss multiplier
Not assessed / absent	1.00
Partial	0.92
Meets minimum	0.80
Enhanced	0.70
Verified	0.60

That should feed both the progression model and the terminal loss threshold, because structural resistance affects whether the event remains a contained fire or becomes a major loss. [\[441\]](#) [\[448\]](#)

External fire resistance scaling

For ExternalMaterialsFireResistant:

Condition	Spread multiplier
Not assessed / absent	1.00

Condition	Spread multiplier
Partial	0.90
Meets minimum	0.78
Enhanced	0.68
Verified	0.60

This is especially important where façade or roof spread could drive floor-to-floor escalation or external flame entry. ^[444] ^[442]

Access and response scaling

For AccessRouteResilient:

Condition	External response multiplier
Not assessed / absent	1.00
Partial	0.92
Meets minimum	0.82
Enhanced	0.72
Verified	0.65

For the height penalty, use a monotone function rather than fixed bins:

$$m_{height} = 1 + 0.02 \times \max(0, \text{storeys} - 3)$$

capped at about 1.60 for very tall or difficult-access assets. That gives you a smooth increase in external-response difficulty as height rises. ^[438] ^[443] ^[444]

Combined response equation

The fire growth update can be written as:

$$I_{t+1} = I_t + \exp(\beta_0 + \beta_E E_i + \beta_F F_i) \times m_{det} \times m_{sup} \times m_{comp} \times m_{firestop} \times m_{extmat}$$

and containment probability as:

$$P(K_t = 1) = \sigma(\kappa_0 + \kappa_D D_i + \kappa_S S_i + \kappa_A A_i - \kappa_H H_i - \kappa_I I_t)$$

where each modifier is mapped from the ranges above. ^[435] ^[437] ^[441]

Governance guidance

These progression and response ranges should be wider than initiation ranges because they are the dominant resilience levers. They also need to be tuned separately from the ignition prior so you do not accidentally double-count the same feature in both frequency and severity. ^[436] ^[449]

Next step

The next deliverable should be a **full parameter table** with every relevant CDM field, the component model it belongs to, the numeric range, and the default multiplier for AA / A / B / NR. That will be the most useful format for implementation.

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Defining conditional dependency logic for fire suppression failure

Yes — this should be implemented as **conditional dependency logic** inside the response and loss models, not as a separate initiation factor. The literature on fire-suppression modelling and fire-resistance failure supports treating suppression failure as a function of detection timing, system availability, fire severity, and compartment conditions. [\[456\]](#) [\[457\]](#) [\[458\]](#)

Dependency structure

The suppression failure node should depend on the following parents:

- `DetectionStatus`.
- `DetectionDelay`.
- `SuppressionSystemsInstalled`.
- `SuppressionSystemCondition`.
- `WaterSupplyAvailability`.
- `PowerAvailability`.
- `FireIntensity`.
- `CompartmentIntegrity`.
- `AccessRouteResilient`.
- `HeightPenalty`.
- `ExternalResponseTime`. [\[459\]](#) [\[460\]](#) [\[456\]](#)

That gives you a complete conditional structure for whether suppression succeeds, partially succeeds, or fails completely. [\[457\]](#) [\[458\]](#)

Suggested state logic

Use a three-way suppression outcome:

1. `SuppressionSuccess`.
2. `SuppressionPartial`.
3. `SuppressionFailure`. [\[460\]](#) [\[457\]](#)

The conditional probability of each state should increase or decrease based on the current state of the fire and the quality of the suppression path. For example, if the fire is detected early, sprinklers are installed and functional, and water/power are available, `SuppressionSuccess` should dominate.

If detection is late, the fire intensity is already high, or the building is tall with poor access, `SuppressionFailure` should rise sharply. ^[458] ^[456] ^[459]

Practical dependency rule

A simple implementation is:

$$P(\text{SuppressionSuccess}) = \sigma(a_0 + a_1D + a_2S + a_3W + a_4P - a_5I - a_6H - a_7T)$$

where:

- D = early detection quality,
- S = suppression system quality,
- W = water supply availability,
- P = power availability,
- I = current fire intensity,
- H = height/access penalty,
- T = response time. ^[456] ^[460] ^[458]

Then define:

$$P(\text{SuppressionFailure}) = 1 - P(\text{SuppressionSuccess}) - P(\text{SuppressionPartial})$$

with `SuppressionPartial` used for cases where the system acts but cannot fully control the fire. ^[457]
^[456]

What “failure” means

Suppression failure should not mean “system absent.” It means the fire has progressed beyond the suppressive capability of the installed and available response path. That can happen because the system is disabled, damaged, lacks water, lacks power, activates too late, or is overwhelmed by fire intensity. This distinction is important for BRI because a building can have installed suppression but still be vulnerable if the response chain is weak. ^[461] ^[459] ^[458] ^[457]

Governance-friendly breakdown

For governance, split suppression dependency into two layers:

- **System readiness layer:** installation, condition, water, power, maintenance.
- **Operational success layer:** detection time, activation timing, intensity, access, brigade intervention. ^[460] ^[456]

This makes validation easier because the first layer is mostly asset-specific, while the second layer is scenario-specific. ^[459] ^[458]

BRD wording

A good BRD requirement would be:

“The suppression failure model shall be conditional on system availability, detection timing, water and power availability, current fire intensity, building access, and external response timing. The model shall distinguish between successful suppression, partial suppression, and complete suppression failure, and shall update these states at each 15-minute interval.” [\[458\]](#) [\[456\]](#) [\[457\]](#)

Next step

The next thing to define is the **explicit conditional table** for suppression failure — i.e. how the probability changes for each combination of detection status, sprinkler status, water/power status, and height/access condition.

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explicit conditional table for suppression failure

Yes — here is a practical **explicit conditional table** for suppression failure that your team can code directly. I’ve kept it simple enough to govern, but structured enough to support Bayesian updating and 15-minute time steps.

Suppression outcome states

Use three outcomes at each step:

- Success.
- Partial.
- Failure.

Interpretation:

- Success means the fire is brought under control within the current interval.
- Partial means the system acts but does not fully contain the fire.
- Failure means the suppression path does not stop escalation.

Parent variables

For the first version, use these parents:

- DetectionStatus = Early, Late, None.
- SprinklerInstalled = Yes, No.
- SprinklerFunctional = Yes, Degraded, No.
- WaterAvailable = Yes, Limited, No.
- PowerAvailable = Yes, No.
- FireIntensity = Low, Medium, High.
- HeightAccess = Low, Medium, High.

- ExternalResponse = Fast, Moderate, Slow.

Conditional table

1. If no sprinkler is installed

Detection	Intensity	Water	Power	Access	Success	Partial	Failure
Early	Low	Yes	Yes	Low	0.45	0.35	0.20
Early	Medium	Yes	Yes	Low	0.30	0.35	0.35
Early	High	Yes	Yes	Low	0.15	0.25	0.60
Late	Low	Yes	Yes	Low	0.25	0.35	0.40
Late	Medium	Yes	Yes	Low	0.15	0.25	0.60
Late	High	Yes	Yes	Low	0.05	0.15	0.80
None	Low	Yes	Yes	Low	0.10	0.25	0.65
None	Medium	Yes	Yes	Low	0.05	0.15	0.80
None	High	Yes	Yes	Low	0.01	0.09	0.90

2. If a sprinkler is installed and functional

Detection	Intensity	Water	Power	Access	Success	Partial	Failure
Early	Low	Yes	Yes	Low	0.85	0.10	0.05
Early	Medium	Yes	Yes	Low	0.70	0.18	0.12
Early	High	Yes	Yes	Low	0.50	0.20	0.30
Late	Low	Yes	Yes	Low	0.70	0.18	0.12
Late	Medium	Yes	Yes	Low	0.50	0.22	0.28
Late	High	Yes	Yes	Low	0.25	0.20	0.55
None	Low	Yes	Yes	Low	0.40	0.25	0.35
None	Medium	Yes	Yes	Low	0.25	0.20	0.55
None	High	Yes	Yes	Low	0.10	0.15	0.75

3. Height and access adjustment

Apply a penalty multiplier to Success and a corresponding increase to Failure:

HeightAccess	Success multiplier	Failure multiplier
Low	1.00	1.00
Medium	0.85	1.15
High	0.65	1.35

This is where upper floors, poor external access, or difficult firefighting geometry should materially weaken suppression. ^[467] ^[468] ^[469]

4. Water and power adjustment

Condition	Success multiplier	Failure multiplier
Water yes, power yes	1.00	1.00
Water limited	0.80	1.20
Water no	0.40	1.60
Power no	0.85	1.15

If both water and power are degraded, apply both penalties multiplicatively.

5. Sprinkler functional adjustment

SprinklerFunctional	Success multiplier	Failure multiplier
Yes	1.00	1.00
Degraded	0.70	1.30
No	0.25	1.75

This lets you distinguish between installed-but-broken and absent systems.

Recommended implementation rule

A clean coding rule is:

1. Select the base row by `DetectionStatus`, `SprinklerInstalled`, and `FireIntensity`.
2. Apply multipliers for `SprinklerFunctional`, `WaterAvailable`, `PowerAvailable`, and `HeightAccess`.
3. Renormalize so the three probabilities sum to 1.
4. Draw the suppression outcome for that 15-minute interval.

Practical defaults

If you want a more compact version for v1, you can collapse the table into a scoring model:

$$\text{SuppressionScore} = 2D + 3S + 2W + 1P - 3I - 2H - 2T$$

Then map the score to probabilities:

- high score $\rightarrow$ Success,
- mid score $\rightarrow$ Partial,
- low score $\rightarrow$ Failure.

That is easier to code, but the table above is better for initial governance because it is more transparent.

BRD wording

A suitable requirement statement is:

“The suppression failure model shall use a conditional probability table based on detection timing, sprinkler installation and functional condition, water and power availability, fire intensity, and access/height penalty. The model shall output suppression success, partial suppression, or suppression failure at each 15-minute interval, and the probabilities shall be renormalized after application of contextual modifiers.”

Next step

The next thing to define is the equivalent conditional table for detection and ignition-to-growth, so the full chain is explicit end to end.

**

OK, we need to define the point where suppression is not working so that $d(\text{intensity})/dt$ starts to go back up. At that point we have uncontrollable fire

Yes — that needs to be an explicit **threshold state** in the model. Once suppression can no longer reduce intensity growth, the process should switch from a containment regime to an uncontrolled-growth regime, and $d(\text{intensity})/dt$ should become positive again. ^[470] ^[471] ^[472]

Threshold state

Define a state called `SuppressionIneffective` or `UncontrollableFire`. The model enters this state when the combined suppression probability falls below a critical cutoff and the fire intensity remains above the system’s effective control capacity for the current interval. That cutoff should depend on detection timing, sprinkler functionality, water availability, access, height, and current fire intensity. ^[471] ^[473] ^[474]

Practical condition

A simple operational rule is:

$$P(\text{SuppressionSuccess}_t) < \theta_s \quad \text{and} \quad I_t > I_s^*$$

where:

- θ_s is the minimum effective suppression probability.
- I_s^* is the intensity threshold beyond which available suppression is no longer effective. ^[475] ^[471]

When that condition holds, set the intensity update to positive growth again:

$$I_{t+1} = I_t + g_{up}(I_t, F_i, E_i) - g_{sup}(t)$$

but with $g_{sup}(t)$ near zero or zero once suppression is ineffective. ^[470] ^[475]

Fire regime logic

You can think of the model as having two regimes:

1. Controlled regime: suppression reduces or flattens intensity growth.
2. Uncontrolled regime: suppression no longer offsets fuel-driven growth, so intensity rises and damage accumulates faster. ^[476] ^[470]

The transition between regimes is the key BRI threshold. It should occur after repeated suppression failure, late detection, or high-intensity escape from the protected compartment. ^[472] ^[471]

Suggested implementation rule

At each 15-minute step:

1. Compute suppression success probability.
2. If success is high, apply a negative or flattening term to intensity growth.
3. If success is partial, apply a weak suppression term.
4. If success is low and intensity exceeds the control threshold, switch to uncontrolled growth.
5. Mark the scenario as ContainmentFailed and propagate to partial or total loss. ^[471] ^[475] ^[470]

In code terms, the intensity update can be:

$$\Delta I_t = \begin{cases} -\alpha \cdot I_t, & \text{if suppression success} \\ 0, & \text{if partial suppression} \\ +\beta \cdot f(F_i, E_i), & \text{if suppression ineffective} \end{cases}$$

with $\beta > 0$. ^[473] ^[475]

BRI meaning

For BRI, this is the point where the fire stops being a manageable event and becomes a resilience failure. It is also the point at which the model should shift from response scoring to loss escalation and total-loss probability. That makes the threshold very useful because it gives you a clean, auditable boundary between contained and uncontrollable fires. ^[477] ^[472] ^[470] ^[471]

BRD wording

A good BRD sentence is:

“The model shall define an uncontrolled-fire threshold. When suppression efficacy falls below the threshold and fire intensity exceeds the available control capacity, the system shall transition from controlled growth to uncontrolled growth and increase the intensity derivative accordingly.” ^[475] ^[470] ^[471]

Next step

The next thing to define is the explicit equations for the controlled and uncontrolled intensity updates, so your team can implement the regime switch directly.

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explicit equations for the controlled and uncontrolled intensity updates

Yes — this should be made explicit as a **regime switch** with two equations: one for controlled fire and one for uncontrolled fire. The model should switch when suppression efficacy drops below a threshold and intensity is high enough that the installed/available response no longer meaningfully offsets growth. ^[487] ^[488] ^[489]

Controlled regime

In the controlled regime, suppression reduces the effective growth rate:

$$\frac{dI_t}{dt} = \alpha_g G_t - \alpha_s S_t$$

where:

- I_t = fire intensity.
- G_t = raw growth driver from fuel, ignition severity, and compartment conditions.
- S_t = suppression effectiveness.
- $\alpha_g, \alpha_s > 0$. ^[490] ^[491]

A practical discrete-time update is:

$$I_{t+1} = I_t + \Delta t (\alpha_g G_t - \alpha_s S_t)$$

with $\Delta t = 15$ minutes in your model. If suppression is strong, the net derivative becomes zero or negative and the fire is contained or decays. ^[488] ^[489] ^[490] ^[487]

Uncontrolled regime

In the uncontrolled regime, suppression is no longer effective enough to offset growth:

$$\frac{dI_t}{dt} = \alpha_g G_t - \alpha_s S_t + \alpha_u U_t$$

where U_t is an escalation term that becomes active once the fire has escaped the containment envelope. A simpler version is: ^[492] ^[493]

$$I_{t+1} = I_t + \Delta t \alpha_u G_t$$

with $\alpha_u > 0$, after which intensity rises monotonically. This is the point at which the fire is effectively uncontrollable. ^[494] ^[491] ^[495] ^[496]

Regime switch condition

Use a binary switch variable Z_t :

$$Z_t = \begin{cases} 1, & \text{if } P(\text{SuppressionSuccess}_t) \geq \theta_s \text{ and } I_t < I^* \\ 0, & \text{otherwise} \end{cases}$$

Then:

$$\frac{dI_t}{dt} = Z_t(\alpha_g G_t - \alpha_s S_t) + (1 - Z_t)(\alpha_u G_t)$$

where I^* is the control-capacity threshold and θ_s is the minimum effective suppression probability. This creates a clear and auditable transition from controlled to uncontrolled behavior.

[489] [495] [497] [487] [488]

Suggested growth driver

For a first version, define:

$$G_t = \exp(\beta_0 + \beta_1 F_i + \beta_2 E_i + \beta_3 M_i)$$

where:

- F_i = fuel/load potential.
- E_i = exposure/ignition severity.
- M_i = material and compartment vulnerability. [498] [491] [490]

That means weak materials and high fuel load make the fire accelerate faster, while good compartmentation and fire resistance reduce the effective growth driver.

Suppression effectiveness term

A practical suppression term is:

$$S_t = p_{det,t} \cdot p_{sup,t} \cdot p_{water,t} \cdot p_{access,t}$$

where each term is a probability or normalized effectiveness score from your detection/response model. That lets suppression degrade continuously before it fully fails, which is more realistic than a hard on/off effect. [499] [488] [489]

BRI interpretation

The moment the model crosses from the controlled regime to the uncontrolled regime is the BRI fire resilience failure point. At that point, the building is no longer preserving containment, so the loss state should begin to escalate toward partial or total loss. [495] [496] [487]

BRD wording

A clean BRD requirement is:

“The fire model shall implement two fire-intensity regimes. In the controlled regime, suppression shall reduce or offset intensity growth; in the uncontrolled regime, the intensity derivative shall

become positive and the fire shall be marked as uncontrollable. The transition shall occur when suppression efficacy falls below a defined threshold and current fire intensity exceeds the available control capacity.”^[488] ^[489] ^[495]

Next step

The next thing to define is the actual parameter values and thresholds for θ_s , I^* , α_g , α_s , and α_u , so the developers can implement the switch consistently.

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actual parameter values and thresholds

Yes — here is a sensible first-pass parameter set for the controlled/uncontrolled fire intensity model. These should be treated as calibration defaults, not immutable constants, but they are good enough to code and test the logic now.^[508] ^[509]

Core thresholds

Use the following starting values:

Parameter	Suggested value	Meaning
θ_s	0.35	Minimum suppression success probability needed to remain in the controlled regime
I_s^*	0.60	Normalized intensity threshold above which suppression is usually no longer effective
I_{contain}	0.40	Below this, containment is likely if suppression is active
I_{partial}	0.40 to 0.60	Damage likely to become partial loss
I_{total}	0.80	Above this, total loss becomes likely if escalation continues

These values align with the idea that fire becomes uncontrollable when the response chain is weak and the intensity is already materially developed.^[509] ^[510] ^[508]

Growth and suppression coefficients

A practical starting set is:

Coefficient	Suggested value	Role
α_g	0.08 to 0.15	Baseline growth sensitivity in the controlled regime
α_s	0.10 to 0.20	Suppression effectiveness in the controlled regime
α_u	0.18 to 0.35	Growth sensitivity in the uncontrolled regime
β_0	-1.5 to -0.5	Baseline growth intercept
β_1	0.6 to 1.2	Fuel/load sensitivity
β_2	0.5 to 1.0	Ignition/exposure severity sensitivity
β_3	0.4 to 0.9	Material and compartment vulnerability sensitivity

These ranges are intentionally broader for the uncontrolled regime because the intensity should rise faster once suppression no longer offsets growth. [\[511\]](#) [\[512\]](#) [\[513\]](#)

Suppression effectiveness scoring

Define suppression effectiveness on a 0 to 1 scale:

$$S_t = p_{det,t} \cdot p_{sup,t} \cdot p_{water,t} \cdot p_{access,t}$$

Use the following benchmark values:

Component	Good	Moderate	Poor
$p_{det,t}$	0.9	0.7	0.4
$p_{sup,t}$	0.9	0.7	0.3
$p_{water,t}$	1.0	0.7	0.3
$p_{access,t}$	1.0	0.75	0.5

The product gives you a natural suppression score that can fall below θ_s and trigger the uncontrolled regime. [\[514\]](#) [\[508\]](#) [\[509\]](#)

Regime-switch rule

The model should switch to uncontrolled growth when both conditions hold:

1. $P(\text{SuppressionSuccess}_t) < 0.35$.
2. $I_t > 0.60$.

That is the point where the building no longer contains the fire, and the intensity derivative should become positive again. [\[515\]](#) [\[516\]](#) [\[517\]](#)

Recommended discrete update

If controlled:

$$I_{t+1} = \max(0, I_t + 0.25 \cdot G_t - 0.20 \cdot S_t)$$

If uncontrolled:

$$I_{t+1} = \min(1, I_t + 0.35 \cdot G_t)$$

where G_t is the raw growth driver normalized to 0–1. [\[518\]](#) [\[519\]](#)

Calibration guidance

These are first-pass values and should be revised by hazard class and building type. For example:

- warehouses and industrial assets may need higher α_g ,
- high-rise assets may need a higher I^* threshold for internal ignition but a lower external-response success rate,
- buildings with verified sprinklers may need lower I_t growth rates and higher $p_{sup,t}$. [\[520\]](#) [\[521\]](#) [\[522\]](#)

BRD wording

A suitable requirement is:

“The model shall define numerical thresholds for suppression effectiveness and fire intensity that govern the transition between controlled and uncontrolled fire regimes. The model shall use calibrated default values for initial implementation and shall permit these values to be versioned and updated by hazard class, building class, and evidence source.” [\[508\]](#) [\[509\]](#) [\[515\]](#)

Next step

The next thing to define is the **full state transition matrix** for the fire progression model, so the equations and the thresholds are tied to explicit probability moves between states.



OK full state transition matrix and I believe the methodology is finalised

Yes — the methodology is now sufficiently defined. Below is the **full state transition matrix** for the fire progression model, using a 15-minute discrete-time Bayesian / Markov-style update and the controlled-to-uncontrolled regime switch. [\[532\]](#) [\[533\]](#) [\[534\]](#)

State set

Use these states:

- S0_IgnitionOrExposure.
- S1_Growth.
- S2_Detected.
- S3_InternalResponse.
- S4_ExternalResponse.
- S5_Contained.
- S6_PartialLoss.
- S7_TotalLoss.
- S8_UncontrollableFire as a regime flag, not an outcome state. [\[533\]](#) [\[535\]](#) [\[536\]](#)

S8_UncontrollableFire is triggered when suppression is ineffective and intensity is increasing again; once active it redirects the flow toward partial or total loss. [\[537\]](#) [\[538\]](#) [\[539\]](#)

Transition matrix

Let the rows be current states and the columns be next states.

From \ To	S1 Growth	S2 Detected	S3 InternalResponse	S4 ExternalResponse	S5 Contained	S6 PartialLoss	S7 TotalLoss
S0 IgnitionOrExposure	p_{01}	p_{02}	0	0	0	0	0
S1 Growth	p_{11}	p_{12}	p_{13}	p_{14}	0	p_{16}	p_{17}
S2 Detected	0	0	p_{23}	p_{24}	p_{25}	p_{26}	p_{27}
S3 InternalResponse	0	0	p_{33}	p_{34}	p_{35}	p_{36}	p_{37}
S4 ExternalResponse	0	0	0	p_{44}	p_{45}	p_{46}	p_{47}
S5 Contained	0	0	0	0	1	0	0
S6 PartialLoss	0	0	0	0	0	1 - p_{67}	p_{67}
S7 TotalLoss	0	0	0	0	0	0	1

All rows sum to 1. Contained and TotalLoss are absorbing states. PartialLoss may either remain partial or escalate to total loss depending on whether the uncontrolled-fire threshold is crossed.
[\[540\]](#) [\[541\]](#) [\[532\]](#) [\[537\]](#)

Transition definitions

1. Ignition or exposure

$$p_{01} = P(\text{Growth} \mid \text{Ignition}) \quad p_{02} = P(\text{Detected} \mid \text{Ignition})$$

At this stage, the fire either develops into growth or is immediately detected. For most scenarios, $p_{01} + p_{02} = 1$. [\[542\]](#) [\[536\]](#)

2. Growth

$$p_{12} = P(\text{Detected} \mid \text{Growth})$$

$$p_{13} = P(\text{InternalResponse} \mid \text{Growth and Detected})$$

$$p_{14} = P(\text{ExternalResponse} \mid \text{Growth and escalation})$$

$$p_{16} = P(\text{PartialLoss} \mid \text{Growth and failed control})$$

$$p_{17} = P(\text{TotalLoss} \mid \text{Growth and failed control})$$

If detection is delayed and fire intensity rises quickly, the path shifts toward partial or total loss.
[\[538\]](#) [\[532\]](#)

3. Detected

From detection, the model branches into internal or external response, or directly into containment if the control is already effective:

$$p_{23} = P(\text{InternalResponse} \mid \text{Detected})$$

$$p_{24} = P(\text{ExternalResponse} \mid \text{Detected})$$

$$p_{25} = P(\text{Contained} \mid \text{Detected})$$

If suppression fails or response is late, the remaining mass flows to partial or total loss. ^{[543] [537] [538]}

4. Internal response

$$p_{35} = P(\text{Contained} \mid \text{InternalResponse success})$$

$$p_{36} = P(\text{PartialLoss} \mid \text{InternalResponse partial success})$$

$$p_{37} = P(\text{TotalLoss} \mid \text{InternalResponse failure})$$

This depends on sprinkler condition, water, power, compartmentation, and current intensity. ^{[544] [537] [543]}

5. External response

$$p_{45} = P(\text{Contained} \mid \text{ExternalResponse success})$$

$$p_{46} = P(\text{PartialLoss} \mid \text{ExternalResponse partial success})$$

$$p_{47} = P(\text{TotalLoss} \mid \text{ExternalResponse failure})$$

This is where height, access, brigade delay, and upper-floor effects should dominate. ^{[535] [545] [546]}

6. Partial loss

$$p_{67} = P(\text{PartialLoss} \rightarrow \text{TotalLoss})$$

This should depend on whether the fire has transitioned into the uncontrolled regime and whether further suppression is effective. ^{[539] [537] [538]}

Regime switch

Define:

$$Z_t = \begin{cases} 1, & \text{controlled regime} \\ 0, & \text{uncontrolled regime} \end{cases}$$

with:

$$Z_t = 1 \text{ if } P(\text{SuppressionSuccess}_t) \geq \theta_s \text{ and } I_t < I^*$$

Otherwise $Z_t = 0$ and the model shifts toward PartialLoss or TotalLoss. ^{[547] [537] [538]}

Probability construction

A convenient implementation is:

$$p_{35} = Z_t \cdot q_{35}$$

$$p_{36} = Z_t \cdot q_{36} + (1 - Z_t) \cdot q'_{36}$$

$$p_{37} = Z_t \cdot q_{37} + (1 - Z_t) \cdot q'_{37}$$

where the q terms are the base outcome probabilities conditional on the current state and the relevant modifiers. That gives you a clean controlled/uncontrolled split while keeping the matrix operationally simple. [\[548\]](#) [\[533\]](#)

Governance note

This is a good final form for the methodology because it separates:

- initiation frequency,
- progression intensity,
- suppression failure,
- loss terminal states. [\[549\]](#) [\[533\]](#) [\[535\]](#)

That separation is exactly what you need for model governance, calibration, and later validation.

Next step

The next thing to draft is the **developer specification**: the exact input nodes, output nodes, and field mapping from the v2 CDM into each matrix transition.

✱✱

great can you create a .md format developer specification.

Below is a ready-to-save .md developer specification. I've structured it for implementation, model governance, and later validation, and it aligns with the modular fire methodology we've defined. [\[557\]](#) [\[558\]](#) [\[559\]](#)

Fire model developer spec

```
# BRI Fire Model Developer Specification
```

```
## 1. Purpose
```

```
This specification defines the fire resilience engine for BRI. The engine shall estimate
```

```
The fire module shall support:
```

- ```
- internal ignition,
- external exposure,
- basement / concealed-space initiation,
```

- process-related ignition,
- upper-floor and high-rise access effects,
- containment success,
- partial loss,
- total loss.

The module shall generate a resilience credit based on simulated outcome frequencies [web:95]

## ## 2. Architecture

The system shall be decomposed into four governed component models:

### 1. **\*\*Initiation Model\*\***

Estimates annual ignition frequency using a Poisson process.

### 2. **\*\*Progression Model\*\***

Estimates fire intensity growth over time and the transition between controlled and uncontrolled fire.

### 3. **\*\*Detection and Response Model\*\***

Estimates detection, internal suppression, and external fire service response.

### 4. **\*\*Loss and Credit Model\*\***

Converts scenario outcomes into contained, partial loss, total loss, and resilience credit.

Each component model shall be independently versioned, validated, and approved [web:95]

## ## 3. Inputs

### ### 3.1 Core property inputs

The model shall ingest the following property features from the CDM:

- `CommercialType`
- `UseClassUK0`
- `BusinessRatesCategory`
- `OccupancyStatus`
- `PropertyCondition`
- `NumberOfStoreys`
- `PropertyPeriod`
- `ConstructionType`
- `FloorType`
- `FloorConstruction`
- `RoofCover`
- `RoofGeometry`
- `RoofFrame`
- `RoofDeck`
- `SoftStory`
- `ShapeIrregularity`
- `BrickVeneer`
- `GlassType`
- `RetrofitYear`
- `HasCrippleWall`
- `BasementPresent`
- `PlantRoomLocation`
- `ServiceCore`
- `EmergencyExits`
- `GoodsLiftCount`
- `GoodsLiftCapacityKg`
- `LoadingBays`

- `AccessRouteResilient`
- `AutomaticDetectionInstalled`
- `SuppressionSystemsInstalled`
- `StructuralFireResistanceAdequate`
- `CompartmentmentsProvided`
- `FireStoppingAtPenetrations`
- `ExternalMaterialsFireResistant`
- `BusinessContinuityPlanInPlace`
- `EmergencyProceduresTested`
- `ClaimsHistory`
- `LastFireDate`
- `FireDamageSeverity`
- `FireHazardClass`
- `BRIFireScore`
- `FireCodes`

### ### 3.2 Optional / derived inputs

The model may derive latent variables such as:

- `ProcessLoadScore`
- `InitiationRiskScore`
- `DetectionQualityScore`
- `SuppressionQualityScore`
- `ContainmentCapacityScore`
- `HeightAccessPenalty`
- `MaterialVulnerabilityScore`

## ## 4. Initiation model

### ### 4.1 Frequency form

Fire ignition frequency shall be modeled as:

$$N_i \sim \text{Poisson}(\lambda_i)$$

where  $(N_i)$  is the number of initiating fire events for property  $(i)$  over the simulation period.

### ### 4.2 Baseline frequency

$$\lambda_i = \lambda_{0,c} \times m_{occ} \times m_{proc} \times m_{cond} \times m_{hist}$$

where:

- $(\lambda_{0,c})$  = baseline annual ignition frequency by commercial class,
- $(m_{occ})$  = occupancy multiplier,
- $(m_{proc})$  = process-load multiplier,
- $(m_{cond})$  = condition multiplier,
- $(m_{hist})$  = history multiplier,
- $(m_{prot})$  = protection-related initiation modifier [web:178][web:170].

### ### 4.3 Recommended default ranges

- `m\_occ`: 0.75 to 1.30
- `m\_proc`: 1.00 to 2.50
- `m\_cond`: 0.85 to 1.40

- `m\_hist`: 0.90 to 1.60
- `m\_prot`: 0.85 to 1.05

### ### 4.4 Output

The initiation model shall output:

- ignition event count,
- initiation class,
- initiation timestamp,
- ignition pathway type.

## ## 5. Progression model

### ### 5.1 State variable

The model shall track normalized fire intensity  $(I_t)$  on a 0 to 1 scale.

### ### 5.2 Controlled regime

If suppression remains effective, intensity shall evolve as:

$$I_{t+1} = \max\left(0, I_t + \Delta t \cdot (\alpha_g G_t - \alpha_s S_t)\right)$$

where:

- $(G_t)$  = growth driver,
- $(S_t)$  = suppression effectiveness,
- $(\alpha_g)$  = growth coefficient,
- $(\alpha_s)$  = suppression coefficient,
- $(\Delta t = 15)$  minutes [web:255][web:258].

### ### 5.3 Uncontrolled regime

If suppression is ineffective, intensity shall evolve as:

$$I_{t+1} = \min\left(1, I_t + \Delta t \cdot \alpha_u G_t\right)$$

where  $(\alpha_u)$  is the uncontrolled growth coefficient [web:247][web:260].

### ### 5.4 Regime switch

The model shall switch from controlled to uncontrolled regime when:

$$P(\text{SuppressionSuccess}_t) < \theta_s \quad \text{and} \quad I_t > I^*$$

with default values:

- $(\theta_s = 0.35)$
- $(I^* = 0.60)$  [web:228][web:237].

### ### 5.5 Default coefficients

- $(\alpha_g = 0.08)$  to  $(0.15)$
- $(\alpha_s = 0.10)$  to  $(0.20)$

-  $\alpha_u = 0.18$  to  $0.35$

### 5.6 Output

The progression model shall output:

- intensity trajectory,
- controlled/uncontrolled regime flag,
- escalation status,
- transition to partial or total loss risk states.

## 6. Detection and response model

### 6.1 Detection probability

The detection model shall estimate:

$$P(D_t=1) = f(\text{AutomaticDetectionInstalled}, \text{OccupancyStatus}, \text{Emergency})$$

### 6.2 Suppression success

The suppression model shall estimate:

$$P(S_t=1) = f(\text{SuppressionSystemsInstalled}, \text{WaterAvailability}, \text{PowerAv})$$

### 6.3 Partial suppression

The model shall support three suppression outcomes:

- success,
- partial,
- failure.

### 6.4 Conditional logic

The suppression model shall transition to failure when available suppression capacity ca

### 6.5 Recommended multipliers

- `AutomaticDetectionInstalled`: 0.65 to 0.95
- `SuppressionSystemsInstalled`: 0.50 to 0.90
- `CompartmentsProvided`: 0.55 to 0.90
- `FireStoppingAtPenetrations`: 0.68 to 0.95
- `StructuralFireResistanceAdequate`: 0.60 to 0.95
- `ExternalMaterialsFireResistant`: 0.60 to 0.95
- `AccessRouteResilient`: 0.65 to 0.95

## 7. State-transition model

### 7.1 States

The model shall support the following states:

- `S0\_IgnitionOrExposure`
- `S1\_Growth`
- `S2\_Detected`
- `S3\_InternalResponse`

- `S4\_ExternalResponse`
- `S5\_Contained`
- `S6\_PartialLoss`
- `S7\_TotalLoss`
- `S8\_UncontrollableFire` (regime flag only) [web:275][web:118].

### ### 7.2 Transition matrix

The model shall use a probabilistic transition matrix updated every 15 minutes. Transiti

### ### 7.3 Absorbing states

`S5\_Contained` and `S7\_TotalLoss` shall be absorbing states.

`S6\_PartialLoss` may escalate to `S7\_TotalLoss` if the uncontrolled-fire regime persists

## ## 8. Loss and credit model

### ### 8.1 Loss classification

The model shall classify each scenario as:

- contained,
- partial loss,
- total loss.

### ### 8.2 Resilience credit

The resilience credit shall be computed as:

$$\backslash[ \text{Loss Frequency} = \frac{N_{\text{partial}} + N_{\text{total}}}{N_{\text{sim}}} \backslash]$$

$$\backslash[ \text{Containment Rate} = \frac{N_{\text{contained}}}{N_{\text{sim}}} \backslash]$$

The BRI fire credit shall be inversely related to loss frequency and positively related

## ## 9. Output data

The model shall return:

- ignition count,
- ignition class,
- state path,
- containment time,
- time-to-overwhelm,
- partial loss indicator,
- total loss indicator,
- loss severity class,
- resilience credit contribution,
- portfolio-level aggregation metrics.

## ## 10. Governance

Each component model shall:

- have a documented purpose,
- define inputs and outputs,
- specify numerical thresholds and ranges,

- be version controlled,
- be independently validated,
- be reproducible from saved configuration,
- be reviewed through a model governance process [web:95][web:292][web:290].

### ## 11. Implementation notes

The model shall be implemented as:

1. a scenario generator,
2. a state updater,
3. a response evaluator,
4. a loss aggregator.

The fire model shall operate on 15-minute intervals and shall support 10,000-scenario Mo

### ## 12. Acceptance criteria

The implementation shall be considered compliant when:

- the initiation model generates ignition counts consistent with Poisson frequency assum
- the progression model switches between controlled and uncontrolled regimes based on th
- the response model outputs success, partial, and failure states,
- the state-transition matrix conserves probability mass,
- the loss model produces stable containment / loss frequencies under repeated runs,
- the final BRI fire credit is reproducible from the same inputs and model version [web:

### ## 13. Future extensions

The architecture shall support later addition of:

- wildfire exposure,
- high-rise-specific fire submodels,
- sprinkler impairment submodels,
- evacuation / ASET-RSET modules,
- insurance-linked loss severity curves [web:118][web:287][web:281].

If you want, I can next turn this into a cleaner v2 internal developer note with:

- parameter tables,
- explicit node definitions,
- and a JSON-style schema for the implementation team.

✱✱

**Yes I need all of the parameter tables, node definitions.  
No need for .json schema.**

Below is a developer-ready **parameter and node specification** for the fire module. It is organized so your team can code the model directly, while keeping the component models separately governable.

## 1. Model nodes

### 1.1 Initiation nodes

- CommercialType.
- OccupancyStatus.
- PropertyCondition.
- PropertyPeriod.
- ConstructionType.
- BasementPresent.
- PlantRoomLocation.
- ServiceCore.
- UseClassUK0.
- BusinessRatesCategory.
- ClaimsHistory.
- LastFireDate.
- FireDamageSeverity.
- FireHazardClass.
- BRIFireScore.

### 1.2 Process-load latent nodes

- ProcessLoadScore.
- InitiationRiskScore.
- HousekeepingScore.
- VacancyRiskScore.

### 1.3 Detection nodes

- AutomaticDetectionInstalled.
- EmergencyProceduresTested.
- OccupancyStatus.
- BRIFireScore.
- DetectionQualityScore.
- DetectionProbability.



## 1.4 Suppression nodes

- SuppressionSystemsInstalled.
- WaterAvailability.
- PowerAvailability.
- SuppressionSystemCondition.
- SuppressionSuccessProbability.
- SuppressionPartialProbability.
- SuppressionFailureProbability.

## 1.5 Progression nodes

- FireIntensity\_t.
- FireGrowthDriver.
- MaterialVulnerabilityScore.
- CompartmentIntegrityScore.
- StructuralFireResistanceAdequate.
- CompartmentsProvided.
- FireStoppingAtPenetrations.
- ExternalMaterialsFireResistant.
- HeightAccessPenalty.
- UncontrollableFireFlag.

## 1.6 External response nodes

- AccessRouteResilient.
- EmergencyResponseTime.
- FireStationDistance.
- NumberOfStoreys.
- FloorLevelMeters.
- ExternalResponseProbability.

## 1.7 Loss nodes

- ContainmentProbability.
- PartialLossProbability.
- TotalLossProbability.
- ContainmentTime.
- TimeToOverwhelm.
- LossState.

- BRIFireCredit.

## 2. Initiation parameter tables

### 2.1 Baseline annual ignition frequency by class

| CommercialType | Baseline $\lambda_0$ |
|----------------|----------------------|
| Office         | 0.020                |
| Retail         | 0.030                |
| Hotel          | 0.035                |
| Leisure        | 0.030                |
| Healthcare     | 0.025                |
| MixedUse       | 0.040                |
| Warehouse      | 0.050                |
| Industrial     | 0.060                |
| Other          | 0.030                |

These are annual event frequencies per asset before modifiers.

### 2.2 Occupancy multiplier

| OccupancyStatus  | Multiplier   |
|------------------|--------------|
| Fully occupied   | 1.00         |
| Partially vacant | 0.90         |
| Vacant           | 0.80 to 0.95 |

Use the lower end only if monitoring/security is good; otherwise use the upper end.

### 2.3 Process-load multiplier

| Derived process class                | Multiplier   |
|--------------------------------------|--------------|
| Low-load office                      | 1.00         |
| Standard retail / office             | 1.10         |
| Hotel / leisure / food service       | 1.20 to 1.35 |
| Warehouse / storage                  | 1.25 to 1.50 |
| Light industrial                     | 1.50 to 1.80 |
| Heavy industrial / process-intensive | 1.80 to 2.50 |

## 2.4 Property condition multiplier

| PropertyCondition | Multiplier |
|-------------------|------------|
| Excellent         | 0.85       |
| Good              | 0.95       |
| Fair              | 1.05       |
| Poor              | 1.20       |
| Very poor         | 1.40       |

## 2.5 Claims / history multiplier

| History state                   | Multiplier   |
|---------------------------------|--------------|
| No prior incidents              | 1.00         |
| Old minor incident              | 1.05         |
| Recent minor incident           | 1.10         |
| Recent moderate/severe incident | 1.25 to 1.60 |

## 2.6 Protection modifier for initiation

| Protection level      | Multiplier |
|-----------------------|------------|
| Not assessed / absent | 1.10       |
| Partial               | 1.00       |
| Meets minimum         | 0.95       |
| Enhanced              | 0.90       |
| Verified              | 0.85       |

## 3. Progression parameter tables

### 3.1 Controlled regime coefficients

| Coefficient | Suggested value |
|-------------|-----------------|
| $\alpha_g$  | 0.08 to 0.15    |
| $\alpha_s$  | 0.10 to 0.20    |
| $\beta_0$   | -1.5 to -0.5    |
| $\beta_1$   | 0.6 to 1.2      |
| $\beta_2$   | 0.5 to 1.0      |
| $\beta_3$   | 0.4 to 0.9      |

### 3.2 Uncontrolled regime coefficients

| Coefficient | Suggested value |
|-------------|-----------------|
| $\alpha_u$  | 0.18 to 0.35    |

### 3.3 Regime thresholds

| Threshold            | Suggested value | Meaning                                                      |
|----------------------|-----------------|--------------------------------------------------------------|
| $\theta_s$           | 0.35            | Minimum suppression success probability to remain controlled |
| $I^*$                | 0.60            | Intensity threshold above which control is usually lost      |
| $I_{\text{contain}}$ | 0.40            | Below this containment is likely                             |
| $I_{\text{partial}}$ | 0.40 to 0.60    | Partial-loss transition band                                 |
| $I_{\text{total}}$   | 0.80            | Total-loss transition band                                   |

### 3.4 Growth driver scaling

| Driver                        | Suggested effect  |
|-------------------------------|-------------------|
| Fuel / load potential         | strong positive   |
| Ignition severity             | moderate positive |
| Material vulnerability        | strong positive   |
| Compartment integrity         | strong negative   |
| Structural fire resistance    | negative          |
| Fire stopping                 | negative          |
| External materials resistance | negative          |

A practical growth driver can be normalized to 0 to 1.

## 4. Detection parameter tables

### 4.1 Detection quality multipliers

| Condition             | Detection multiplier |
|-----------------------|----------------------|
| Not assessed / absent | 1.00                 |
| Partial               | 0.90                 |
| Meets minimum         | 0.80                 |
| Enhanced              | 0.72                 |
| Verified              | 0.65                 |

## 4.2 Detection probability inputs

| Node                        | Direction         |
|-----------------------------|-------------------|
| AutomaticDetectionInstalled | positive          |
| EmergencyProceduresTested   | positive          |
| OccupancyStatus             | mixed             |
| BRIFireScore                | positive          |
| Vacancy / low activity      | negative or mixed |

## 5. Suppression parameter tables

### 5.1 Suppression system status multiplier

| Condition             | Suppression multiplier |
|-----------------------|------------------------|
| Not assessed / absent | 1.00                   |
| Partial               | 0.85                   |
| Meets minimum         | 0.75                   |
| Enhanced              | 0.62                   |
| Verified              | 0.50                   |

### 5.2 Functional condition multiplier

| Sprinkler / suppression condition | Multiplier |
|-----------------------------------|------------|
| Yes / functional                  | 1.00       |
| Degraded                          | 0.70       |
| No / failed                       | 0.25       |

### 5.3 Water / power multipliers

| Condition          | Water multiplier | Power multiplier |
|--------------------|------------------|------------------|
| Good               | 1.00             | 1.00             |
| Limited / degraded | 0.80             | 0.85             |
| Absent             | 0.40             | 0.60             |

### 5.4 Suppression success base table

| Detection | Intensity | Base success |
|-----------|-----------|--------------|
| Early     | Low       | 0.85         |
| Early     | Medium    | 0.70         |
| Early     | High      | 0.50         |

| Detection | Intensity | Base success |
|-----------|-----------|--------------|
| Late      | Low       | 0.70         |
| Late      | Medium    | 0.50         |
| Late      | High      | 0.25         |
| None      | Low       | 0.40         |
| None      | Medium    | 0.25         |
| None      | High      | 0.10         |

## 5.5 Suppression failure base table

| Detection | Intensity | Base failure |
|-----------|-----------|--------------|
| Early     | Low       | 0.05         |
| Early     | Medium    | 0.12         |
| Early     | High      | 0.30         |
| Late      | Low       | 0.12         |
| Late      | Medium    | 0.28         |
| Late      | High      | 0.55         |
| None      | Low       | 0.35         |
| None      | Medium    | 0.55         |
| None      | High      | 0.75         |

## 6. External response parameter tables

### 6.1 Access multiplier

| AccessRouteResilient  | External response multiplier |
|-----------------------|------------------------------|
| Not assessed / absent | 1.00                         |
| Partial               | 0.92                         |
| Meets minimum         | 0.82                         |
| Enhanced              | 0.72                         |
| Verified              | 0.65                         |

### 6.2 Height penalty

Use:

$$m_{height} = 1 + 0.02 \times \max(0, \text{NumberOfStoreys} - 3)$$

cap at 1.60.

### 6.3 Emergency response time penalty

| EmergencyResponseTime | Multiplier |
|-----------------------|------------|
| $\leq 5$ min          | 0.90       |
| 6 to 10 min           | 1.00       |
| 11 to 15 min          | 1.15       |
| $> 15$ min            | 1.30       |

### 6.4 External response base table

| Access   | Height | Base external success |
|----------|--------|-----------------------|
| Good     | Low    | 0.85                  |
| Good     | Medium | 0.75                  |
| Good     | High   | 0.60                  |
| Moderate | Low    | 0.70                  |
| Moderate | Medium | 0.55                  |
| Moderate | High   | 0.40                  |
| Poor     | Low    | 0.50                  |
| Poor     | Medium | 0.35                  |
| Poor     | High   | 0.20                  |

## 7. State transition nodes and outputs

### 7.1 Transition inputs

- IgnitionClass.
- Detected.
- SuppressionSuccessProbability.
- SuppressionPartialProbability.
- SuppressionFailureProbability.
- ExternalResponseProbability.
- FireIntensity\_t.
- UncontrollableFireFlag.

### 7.2 Transition outputs

- State\_t+1.
- FireIntensity\_t+1.
- ContainmentProbability.
- PartialLossProbability.

- TotalLossProbability.

## 8. Loss model parameters

### 8.1 Loss state thresholds

| Threshold    | Value                  |
|--------------|------------------------|
| Contained    | 0.00 to 0.40 intensity |
| Partial loss | 0.40 to 0.80 intensity |
| Total loss   | 0.80 to 1.00 intensity |

### 8.2 BRI fire credit definition

$$\text{BRIFireCredit} = 1 - \frac{N_{\text{partial}} + N_{\text{total}}}{N_{\text{sim}}}$$

Optionally report:

- ContainmentRate =  $N_{\text{contained}} / N_{\text{sim}}$
- PartialLossRate =  $N_{\text{partial}} / N_{\text{sim}}$
- TotalLossRate =  $N_{\text{total}} / N_{\text{sim}}$

## 9. Recommended default priors

### 9.1 Suggested defaults by component

- CommercialType sets the initial frequency prior.
- OccupancyStatus modifies initiation and detection.
- PropertyCondition modifies initiation and progression.
- AutomaticDetectionInstalled modifies detection.
- SuppressionSystemsInstalled modifies suppression.
- CompartmentsProvided modifies progression and containment.
- StructuralFireResistanceAdequate modifies progression and loss.
- AccessRouteResilient modifies external response.
- ClaimsHistory modifies all priors modestly.

## 10. Implementation order

1. Build initiation priors.
2. Build detection and suppression tables.
3. Build controlled/uncontrolled progression equations.
4. Build state transition matrix.
5. Build loss aggregation and BRI credit.



6. Add calibration hooks for evidence updates.

## 11. Governance notes

- Do not let initiation modifiers dominate progression modifiers.
- Do not let one field affect more than two component models unless physically justified.
- Version all thresholds and coefficients.
- Keep the initial coefficient set narrow enough for review, but wide enough to discriminate high-quality and low-quality assets.

If you want, I can next convert this into a **\*\*single consolidated master table\*\*** with columns

``Node | Component Model | Field Type | Default | Range | Direction | Notes``

That is usually the easiest format for developers and model governance together.

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